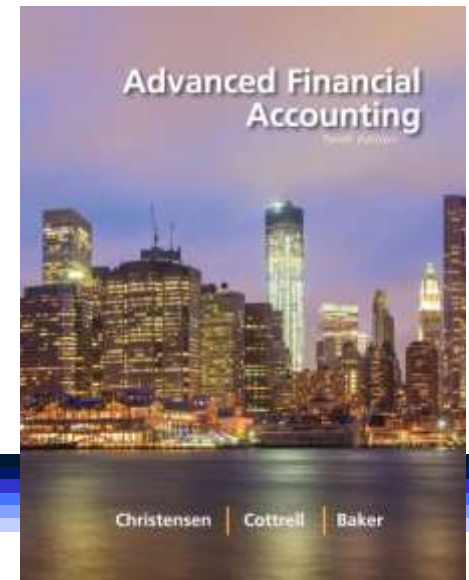


Chapter 6

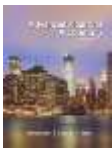


Intercompany Inventory Transactions



Learning Objective 6-1

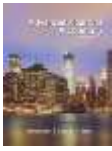
Understand and explain
intercompany transfers and
why they must be
eliminated.



Example 1:

Sale from Parent to Sub to Outsider

- ◆ Parent has 19 subsidiaries.
- ◆ Parent has received a \$1 order from an outsider.
- ◆ Parent sells inventory to Sub 1 for \$1.
 - Sub 1 sells the inventory to Sub 2 for \$1.
 - Sub 2 sells the inventory to Sub 3 for \$1.
 - The inventory is sold from one sub to another until Sub 19 sells it to the outsider for \$1.
- ◆ The parent and each sub reports sales of \$1.
- ◆ From a consolidated standpoint, what is the total amount of sales?



Example 2: Sale from Parent to Sub, But Not Yet to an Outsider

- ◆ Sleazy Parent Company has one sub.
- ◆ Sleazy Parent is preparing for an IPO.
- ◆ Sleazy Parent owns lots of obsolete inventory which it cannot sell.
- ◆ Sleazy Parent sells the obsolete inventory (costing \$1,000) to its sub for \$100,000.
- ◆ Sleazy Sub now holds the inventory.
- ◆ Without any adjustment, what items in Sleazy's consolidated financial statements will be misstated?



Learning Objective 6-2

Understand and explain concepts associated with inventory transfers and transfer pricing.



Issue #1: Eliminate Intercompany Transfers?

◆ Whether to Eliminate Intercompany Transactions in Consolidation:

- No controversy—they *must* be eliminated.
- Not eliminating them would cause two problems:
 - Meaningless **double-counting** of
 1. sales, and
 2. expenses
 - Potential to **manipulate income**.





The Substance of Inventory Transfers

◆ The CONSOLIDATED Perspective:

- Merely the physical movement of inventory from one location to another location.
- Similar to the movement of inventory from one division to another division.
- Not a bona fide transaction.





Issue #2: Eliminate Income Tax Effects?

- ◆ Income taxes play a major role in intercompany sales and transfer pricing decisions.
- ◆ Income taxes on the selling entity's unrealized gross profit must also be eliminated.
- ◆ In this chapter:
 - No income tax entries are required.
 - Because we assume that the tax effects have already been recorded in the parent's or the subsidiary's general ledger.



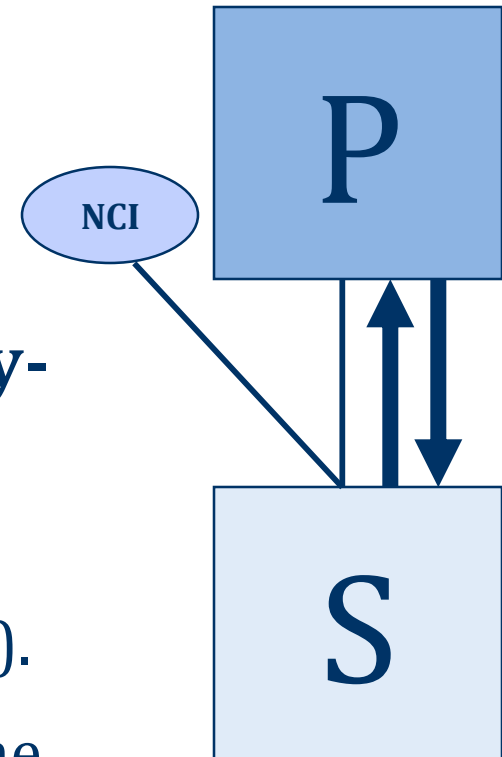
Issue #3: Whether To Eliminate All or Some?

◆ Downstream sales to a partially-owned subsidiary:

- Entire profit accrues to the parent; thus, sharing is not appropriate.

◆ Upstream sales from a partially-owned subsidiary:

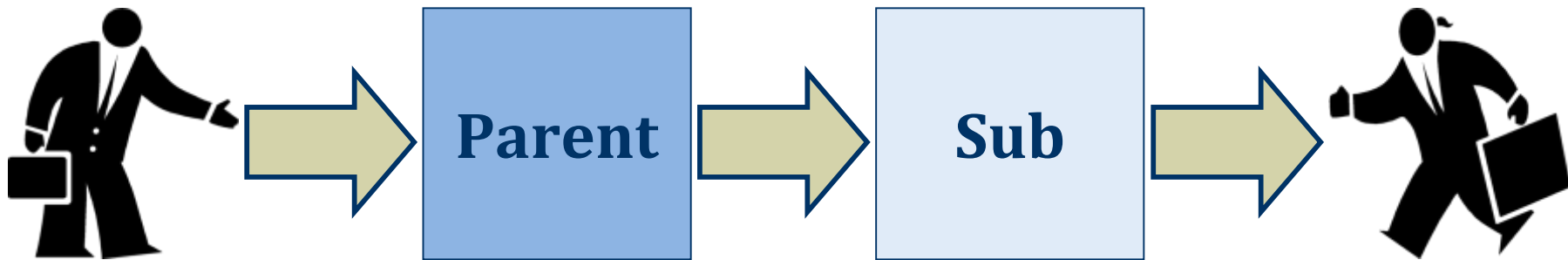
- Must share deferral with the NCI shareholders (if amount is material).
- Because S profits are shared with the NCI shareholders.





Inventory Transfers: What is “Realization”?

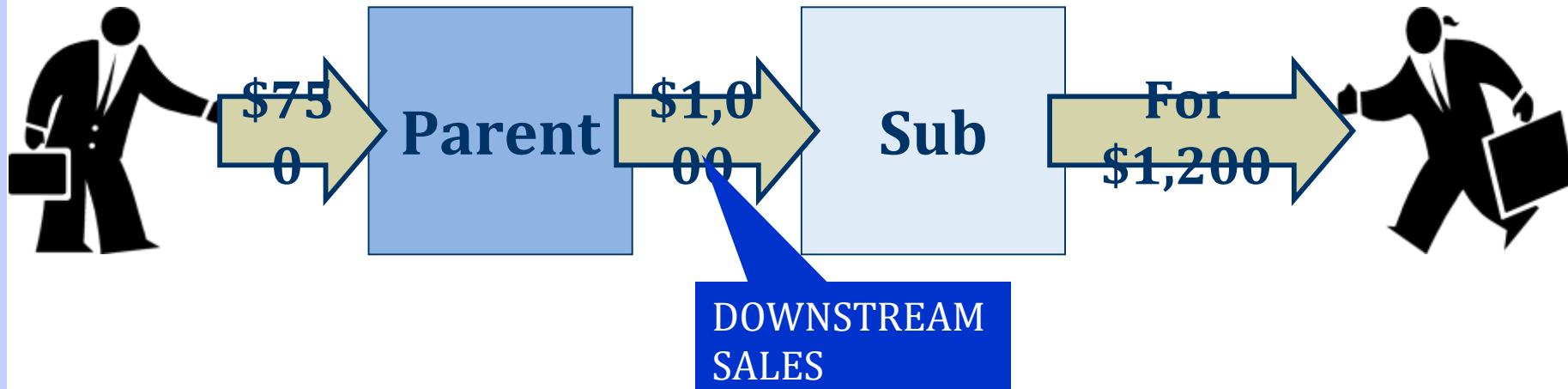
- ◆ Realization for consolidated reporting purposes:
 - Depends on whether the BUYER has resold the inventory to an outside unaffiliated customer.





Review: Two Types of Transfers #1

■ Parent-to-sub-to-outsider

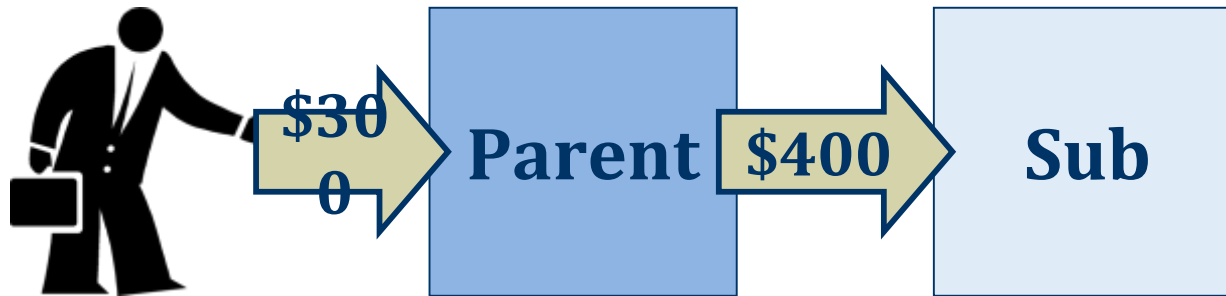


profit from sales has been
REALIZED

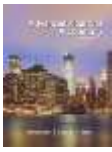


Review: Two Types of Transfers #2

- Parent-to-sub-not-yet-to-outsider

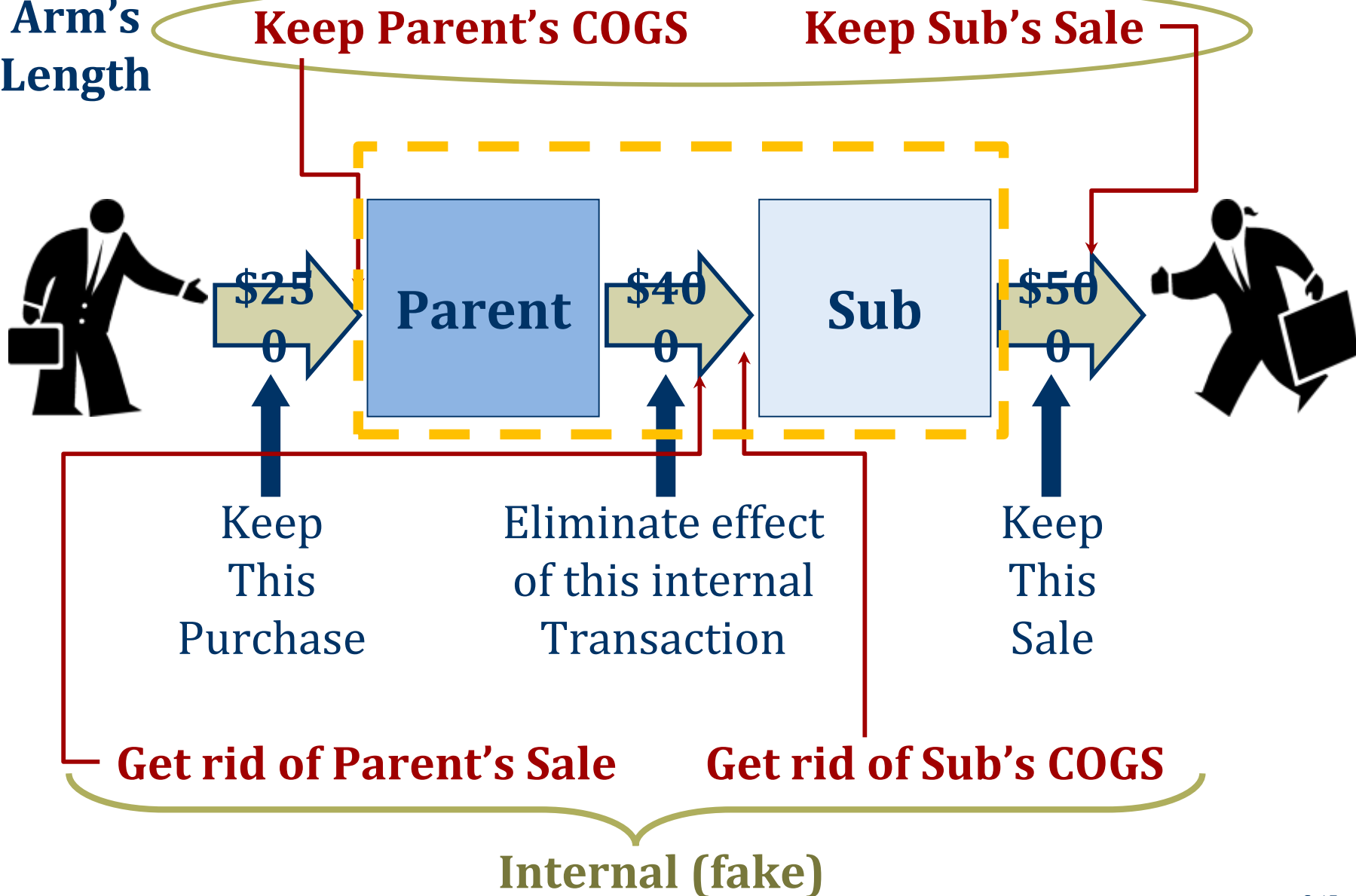


profit from sales has NOT
been REALIZED



(a) Sale from Parent to Sub to Outsider

Arm's
Length





(a) Sale from Parent to Sub to Outsider

Which transactions are legitimate?

Parent's sale to Sub:

Parent:

Cash 400

Sales

400

COGS 250

Sub Inventory

Inventory 250

Cash

Cancel!



Cancel!



Sub's sale to Outsider:

Sub:

Cash 500

Sales

500

COGS 400

Inventory

Inventory 400

Reverse the rest!



To eliminate sale from Parent to Sub to Outsider:

Sales (parent to sub) 400

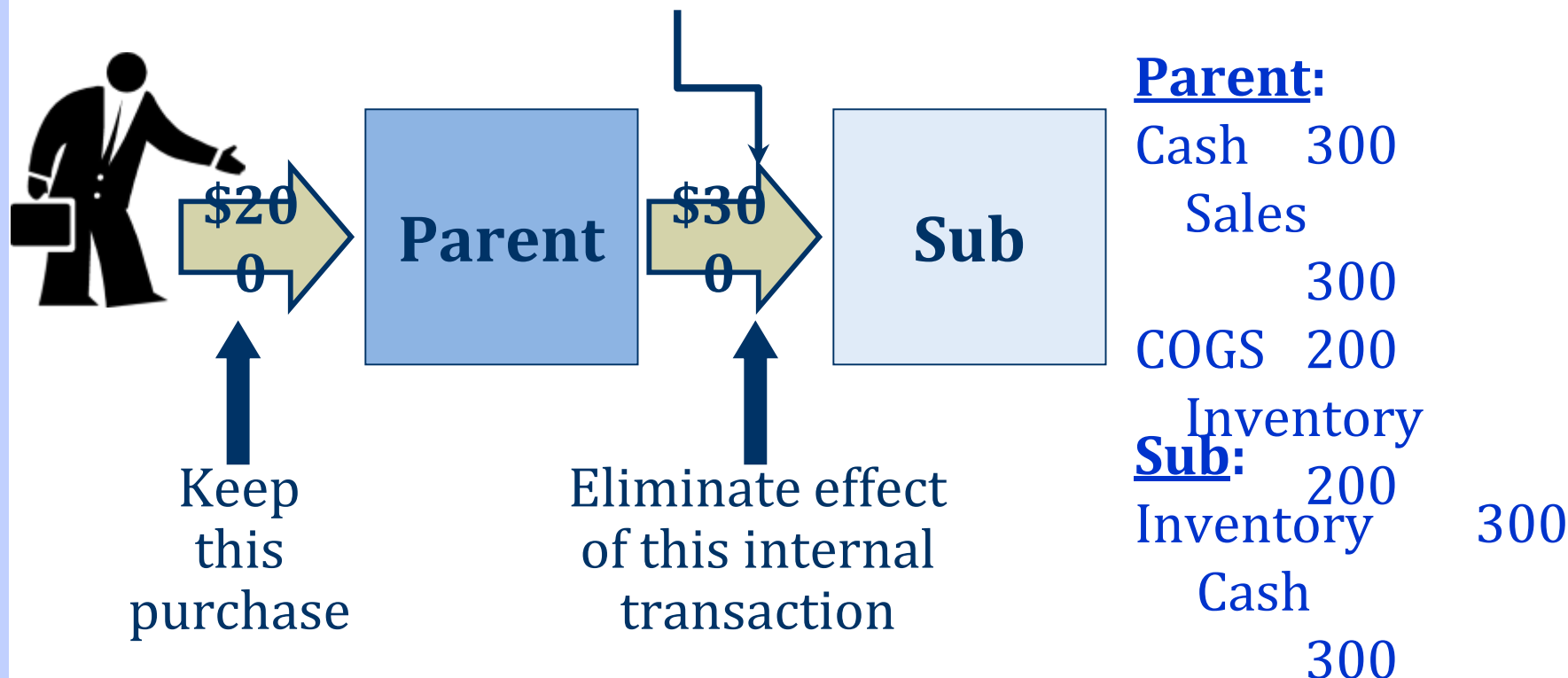
Cost of Goods Sold (to outsider)

400



(b) Sale From Parent to Sub (Not Outside)

Is this a legitimate arm's length transaction?



Summary of the Transaction:

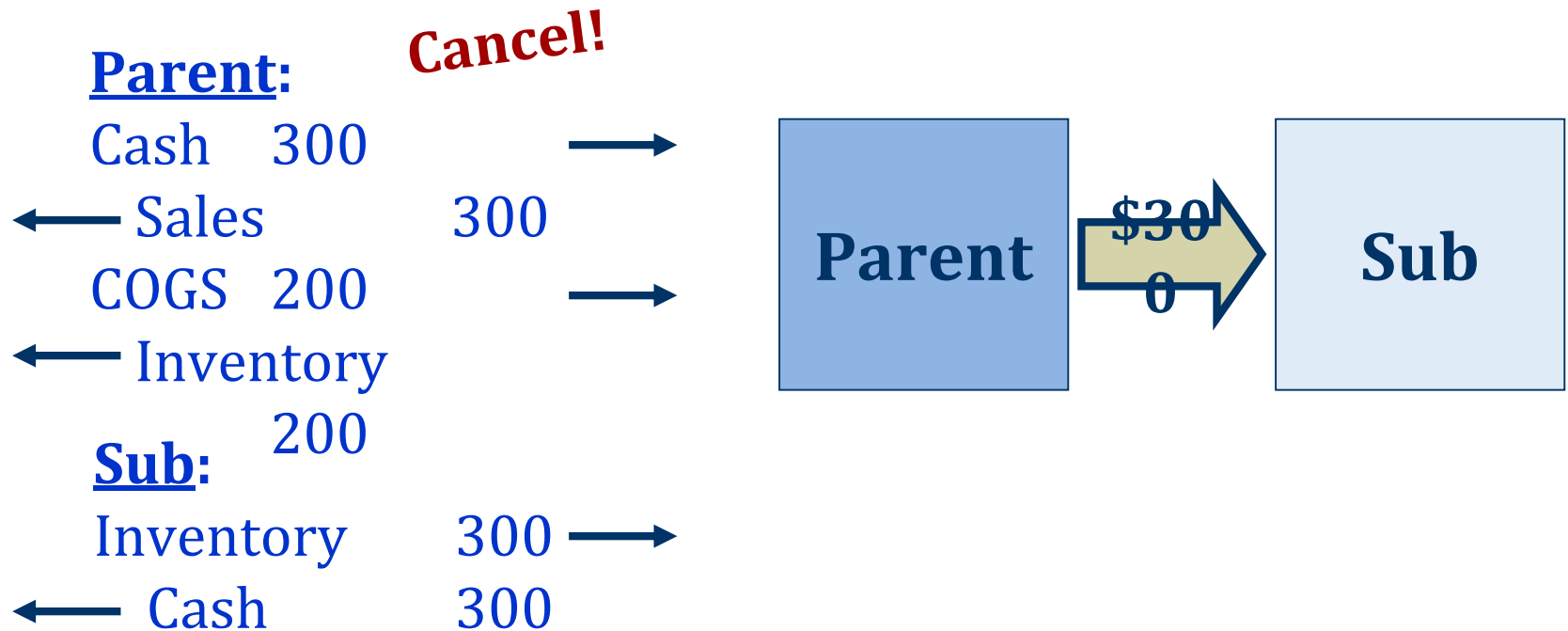
- Parent purchased inventory for \$200.
- Parent sold the inventory to a Sub for \$300.

Reverse the entries made by the parent and sub.



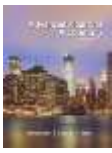
(b) Sale From Parent to Sub (Not Outside)

Reverse the entries made by the parent and sub.



To eliminate sale from Parent to Sub, not yet to Outsider:

Sales	300	
Cost of Goods Sold		200
(net) Inventory	100	



Agreement between Parent Company and Consolidated Financial Statements

- ◆ Under the fully adjusted equity method,
 - the parent company's financial statements should report the same net income and retained earnings amounts as appear in the consolidated statements.
- ◆ Therefore, we
 - record and **equity method adjustment** on the parent's books **to defer unrealized gross profit**, and
 - prepare consolidation **worksheet elimination entries to avoid double counting in the income statement and overstating inventory.**



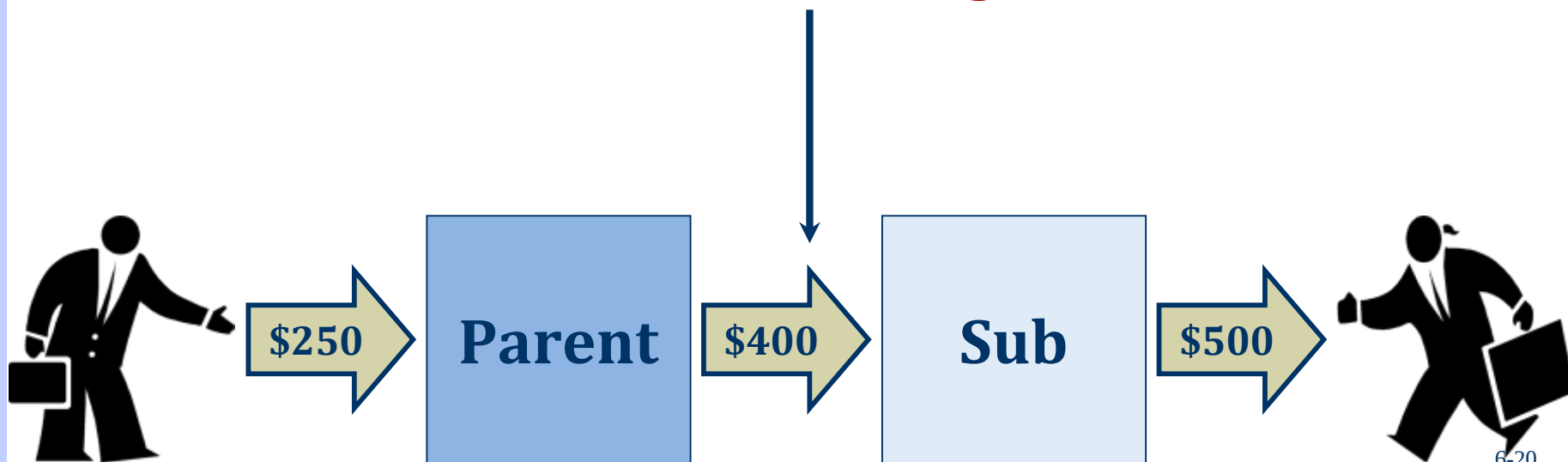
Big Picture—Elimination entry: Sale From Parent to Sub to Outsider

To eliminate sale from Parent to Sub to Outsider:

Sales (Parent) 400

Cost of Goods Sold (Sub) 400

Get rid of the non-arm's-length transaction!





Big Picture—Elimination entry: Sale From Parent to Sub (not yet sold outside)

Reverse the entire transaction!

To eliminate sale from Parent to Sub, not yet to Outsider:

Sales	400		
Cost of Goods Sold			250
Inventory		150	

Equity Method Entry:

Income from Sub	150	
Investment in Sub		150

Sales	\$400
Cost of sales	250
Gross profit	\$ 150

Parent's gross profit is overstated by \$150

Sub's inventory is overstated by \$150



\$250

Parent

\$400

Sub



Fully-adjusted Equity Method Adjustment

- ◆ Parent companies have to adjust their equity method investment accounts for certain transactions.
- ◆ At this point, let's just consider one:
 - Sale from parent to sub, but not yet sold to an outsider.
 - It represents “fake profit” that hasn't really been realized in an arm's-length transaction.
- ◆ Both the balance sheet and income statement accounts need to be adjusted.
- ◆ This is a REAL journal entry, not a consolidation worksheet entry!



Equity Method Adjustment Example



Summary of the Transaction:

- Parent purchased inventory for \$500.
- Parent sold the inventory to a Sub for \$600.

Equity Method Entry:

Income from Sub	100	
Investment in Sub		100

- ◆ The Parent recognized \$100 of “fake” gross profit!
- ◆ The Parent should have transferred the inventory at cost.
- ◆ This profit is not from a transaction with an arm’s-length independent party.



Let's work through an example:

- ◆ Assume Parent Co. owns 100% of Sub Co.
- ◆ The following intercompany transactions occurred during the year:
 - Parent made a sale to Sub for \$400 cash. The inventory had originally cost Parent \$250. Sub then sold that same inventory to an outsider for \$500.
 - Parent made a sale to Sub for \$300 cash. The inventory had originally cost Parent \$200. Sub has not yet sold that same inventory to an outsider.
- ◆ What consolidation worksheet entries would you make?



Group Practice

- ◆ Assume Parent Co. owns 100% of Sub Co.
- ◆ The following intercompany transactions occurred during the year:
 - Parent made a sale to Sub for \$200 cash. The inventory had originally cost Parent \$120. Sub then sold that same inventory to an outsider for \$300.
 - Parent made a sale to Sub for \$300 cash. The inventory had originally cost Parent \$180. Sub has not yet sold that same inventory to an outsider. (Don't forget equity method entry!)



Consolidation Entries

To eliminate sale from Parent to Sub to Outsider:

Sales	200	
	Cost of Goods Sold	200



Consolidation Entries

To eliminate sale from Parent to Sub to Outsider:

Sales	200	
Cost of Goods Sold		200

To eliminate sale from Parent to Sub, not yet to Outsider:

Sales	300	
Cost of Goods Sold		180
Inventory	120	

To correct inventory value



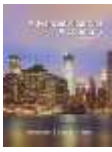
Equity Method Entry:

Income from Sub	120	
Investment in Sub		120

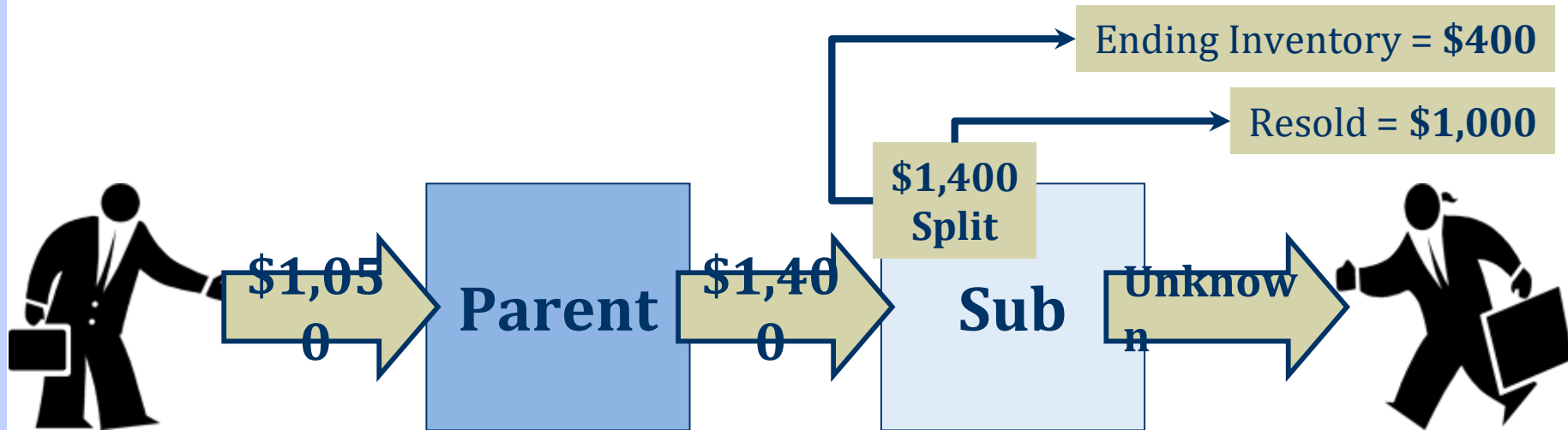


Fully-adjusted Equity Method Adjustment

- ◆ Don't forget that one of the desirable properties of using the equity method is that **the parent's net income should be equal to the consolidated net income.**
- ◆ If you only adjust for unrealized deferred profit in the consolidation, the consolidated net income will be different from the parent's income!

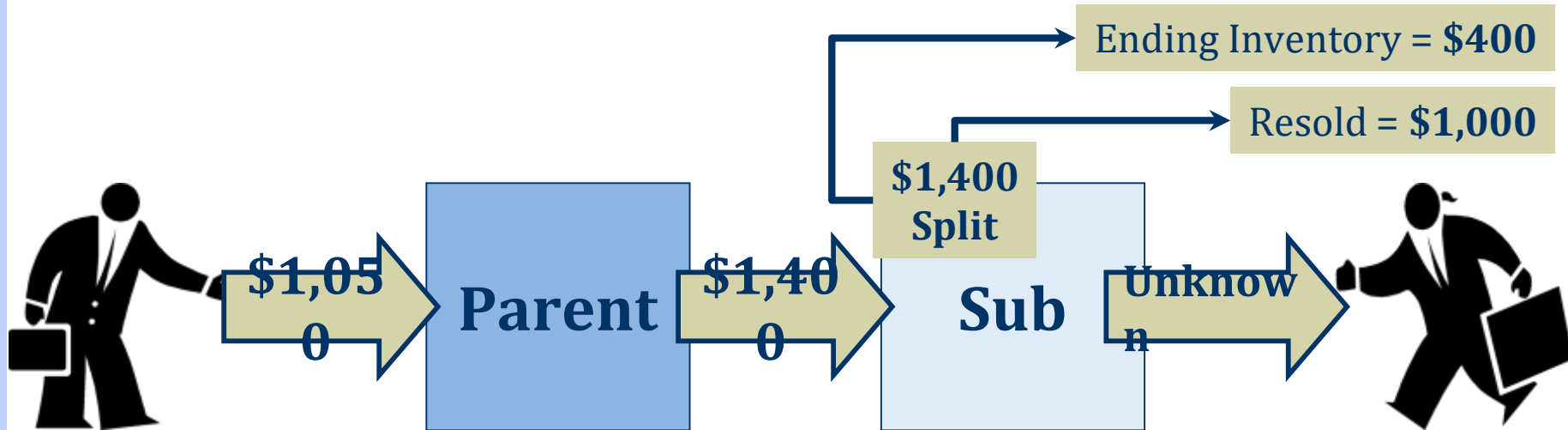


Understanding Inventory Transfers: Map it out





Understanding Inventory Transfers: Map it out



	What happened to it?		
	Total Interco Sales	Resold	On hand
Sales	1,400	1,000	400
– COGS	1,050	750	300
Gross Profit	350	250	100
Gross Profit %	25%		

Splits out parent's numbers.



Inventory Transfers: Terminology

		What happened to it?		
		Total Interco Sales	Resold	On hand
Transfer Price ➡	Sales	1,400	1,000	400
Cost ➡	– COGS	1,050	750	300
Markup ➡	Gross Profit	350	250	100
Markup on Transfer Price ➡	Gross Profit %	25%		

**Watch out for terminology like
“mark-up based on cost”!**



Calculating Unrealized Gross Profit

- ◆ Amounts that will **always** be known (given):

	Total	Resold	On hand
Sales (NEW basis)	1,000		200
– Cost of sales (OLD basis)	600		
Gross Profit	400		
Gross Profit %	40%		

CRITICAL ASSUMPTION:

- ◆ The gross profit percentage derivable from the **total column** applies to both (1) the inventory that has been **resold** AND (2) the inventory that is **still on hand**.



Calculating Unrealized Gross Profit

◆ Completed Analysis:

	Total	Resold	On hand
Sales (NEW basis)	1,000	800	200
– Cost of sales (OLD basis)	600	480	120
Gross Profit	400	320	80
Gross Profit %	40%		

Diagram showing the breakdown of Gross Profit into Realized and Unrealized components:

- The **320** in the Resold column is labeled **Realized** (in blue).
- The **80** in the On hand column is labeled **Unrealized** (in red).
- Arrows indicate that the 40% Gross Profit % applies to both the Resold and On hand columns.

- ◆ The Inventory/COGS Change in Basis Elimination Entry is derived from this analysis.
- ◆ $\text{Unrealized profit} = \text{Inventory on hand} \times \text{GP}\%$
 $= \$200 \times 40\% = \80



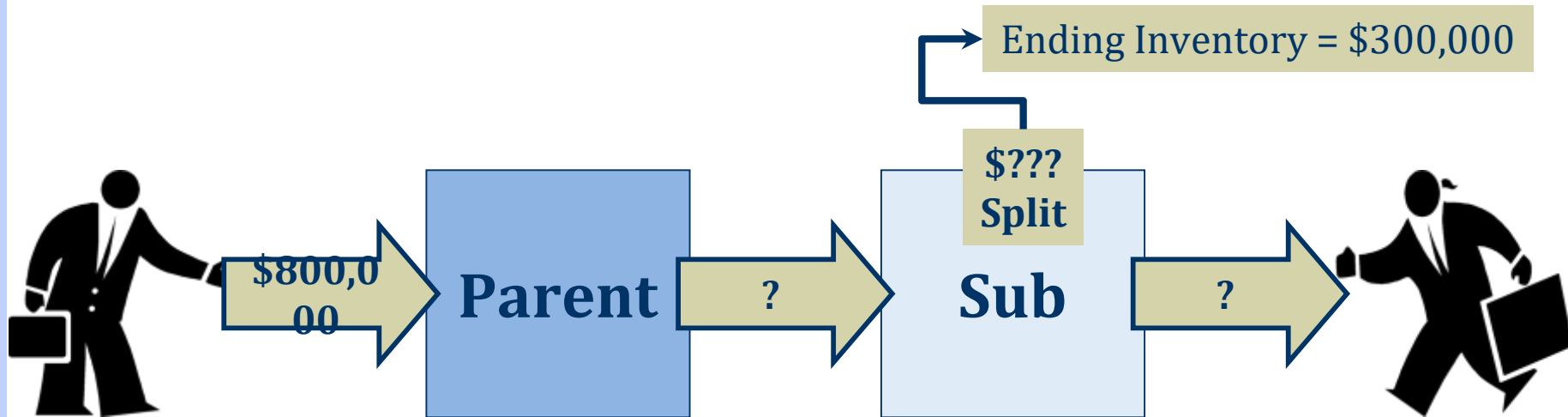
Practice Quiz Question #2

For 20X8, Pete reported intercompany cost of sales of \$800,000 (markup is 20% of *transfer price*) to Sampras, which reported \$300,000 of intercompany acquired inventory at 12/31/X8. The unrealized profit at 12/31/X8 is

- a. \$40,000.
- b. \$48,000.
- c. \$60,000.
- d. \$75,000.
- e. None of the above.



Practice Quiz Question #2 Solution

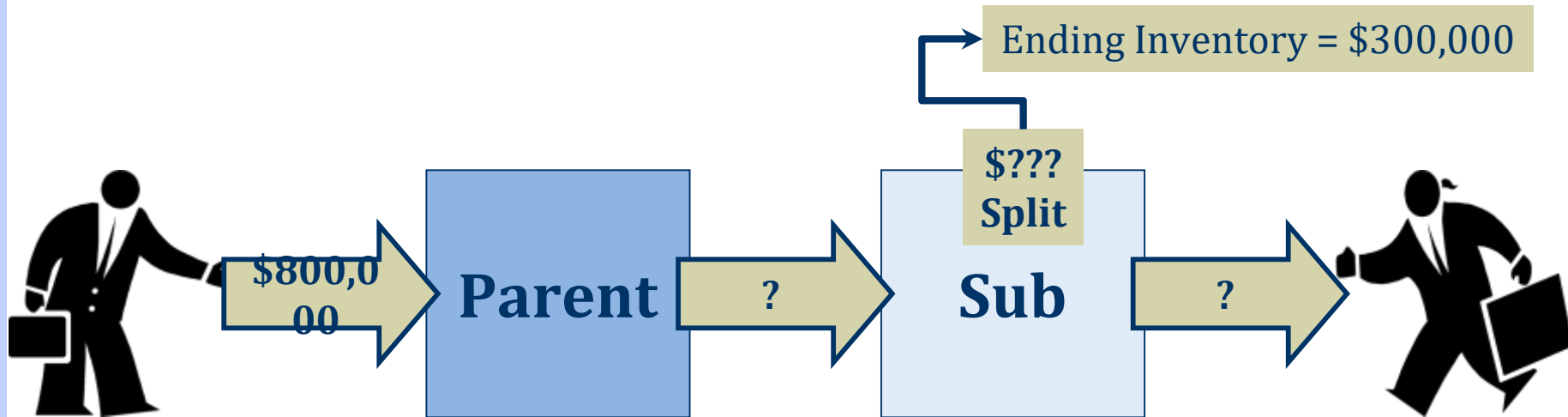


What happened to it?

	Total Interco Sales	Resold	On hand
Sales			300,000
– COGS	800,000		
Gross Profit			?
Gross Profit %	20%		



Practice Quiz Question #2 Solution

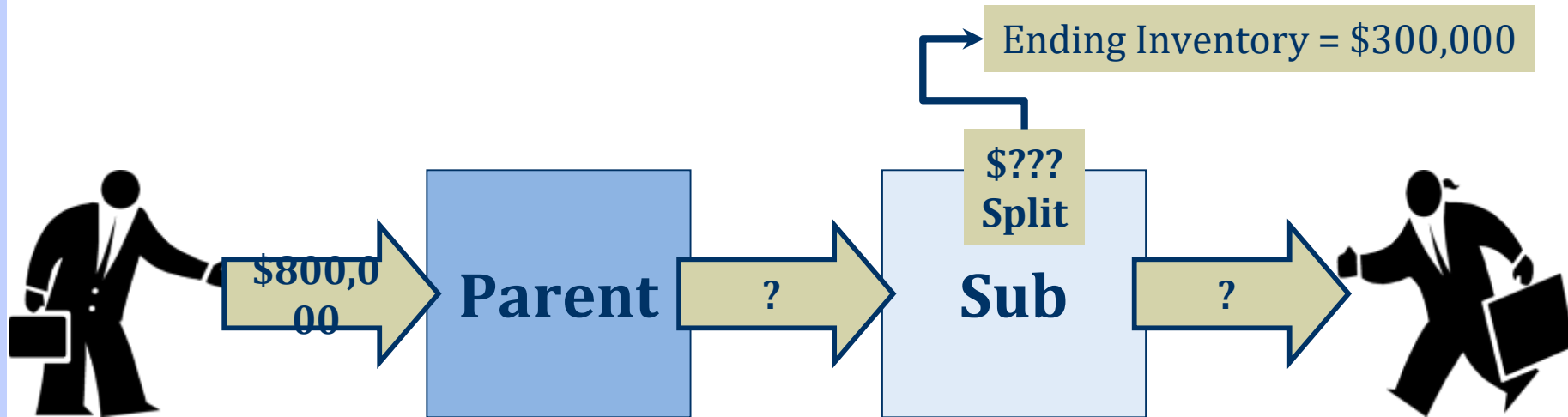


What happened to it?

	Total Interco Sales	Resold	On hand
Sales			300,000
– COGS	800,000		
Gross Profit			60,000
Gross Profit %	20%		



Practice Quiz Question #2 Solution

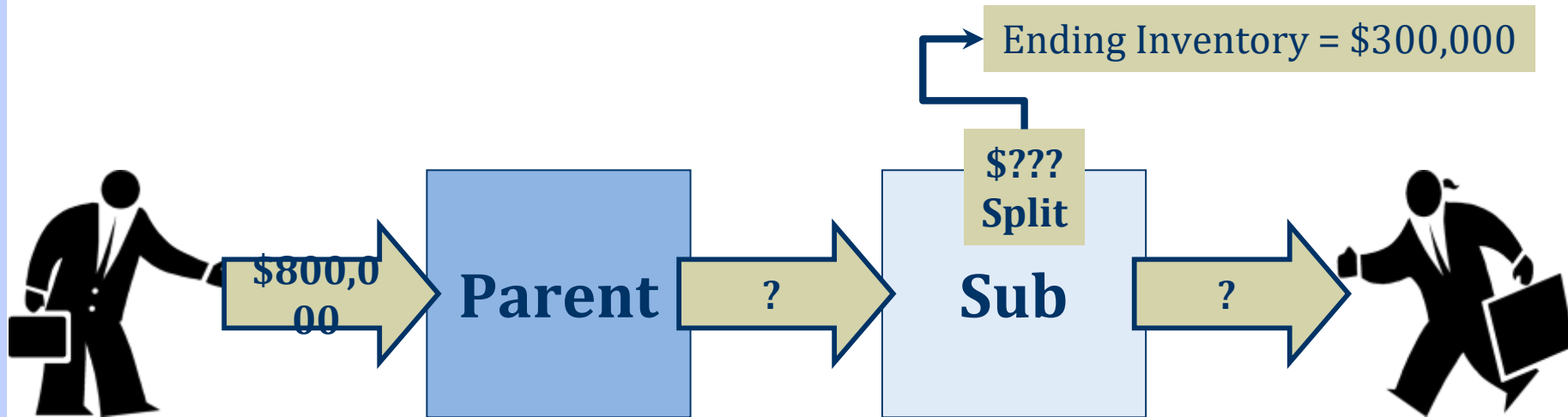


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	\$		300,000
– COGS	800,000		
Gross Profit	.2 \$		60,000
Gross Profit %	20%		



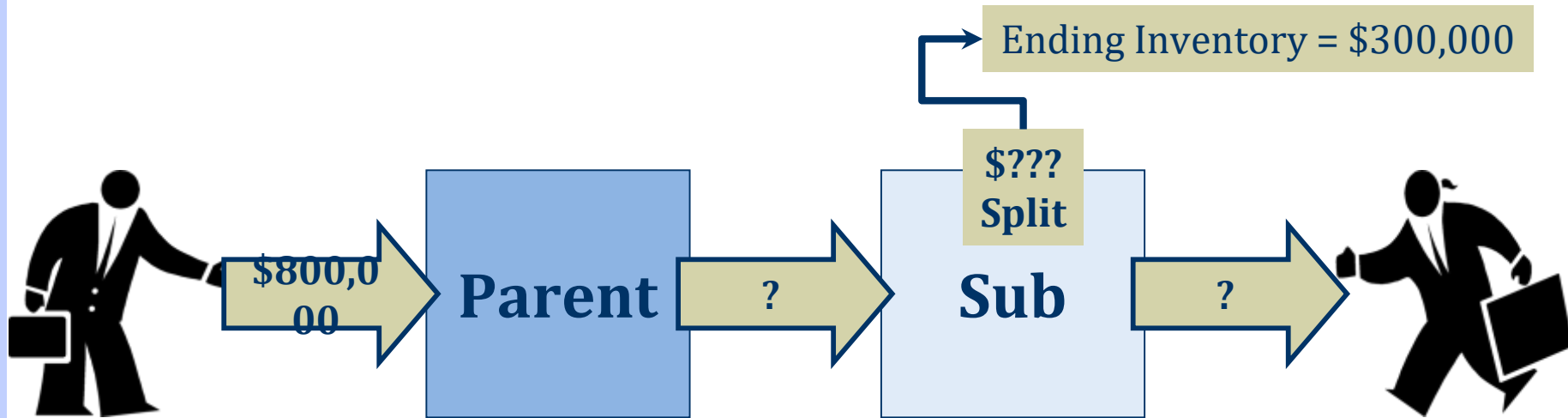
Practice Quiz Question #2 *Solution*



$$\begin{aligned} S - 800,000 &= .2 S \\ .8 S &= 800,000 \\ S &= 800,000 / .8 = 1,000,000 \end{aligned}$$



Practice Quiz Question #2 Solution

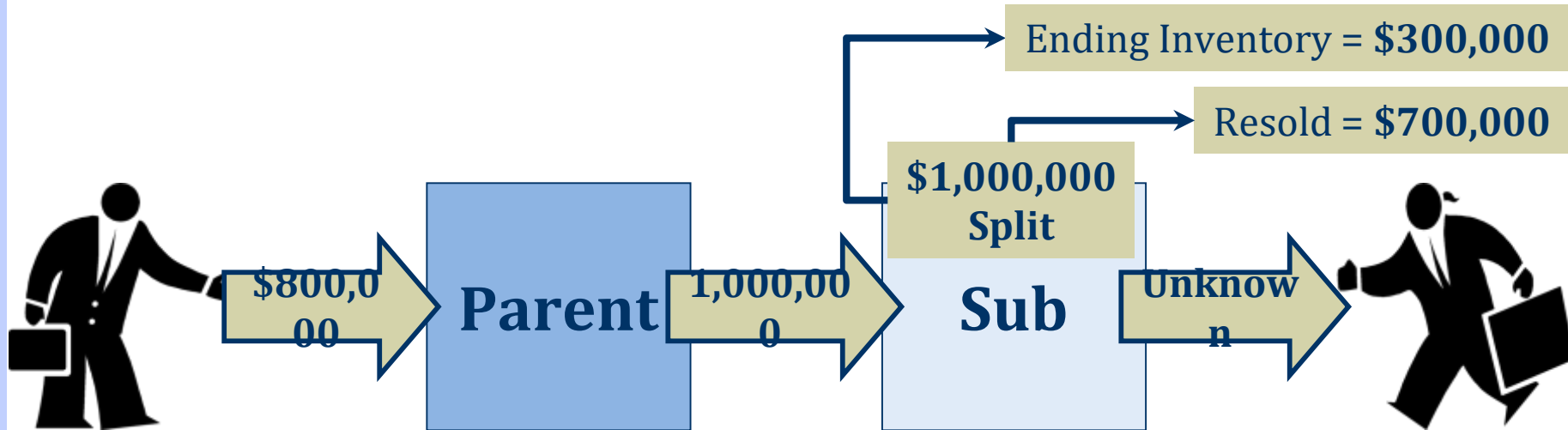


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,000,000		300,000
– COGS	800,000		
Gross Profit	200,000		60,000
Gross Profit %	20%		



Practice Quiz Question #2 Solution



What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,000,000	700,000	300,000
– COGS	800,000	560,000	240,000
Gross Profit	200,000	140,000	60,000
Gross Profit %	20%		



Practice Quiz Question #2 *Solution*

For 20X8, Pete reported intercompany cost of sales of \$800,000 (markup is 20% of *transfer price*) to Sampras, which reported \$300,000 of intercompany acquired inventory at 12/31/X8. The unrealized profit at 12/31/X8 is

- a. \$40,000.
- b. \$48,000.
- c. \$60,000 (\$300,000 EI x 0.20 GP%).
- d. \$75,000.
- e. None of the above.



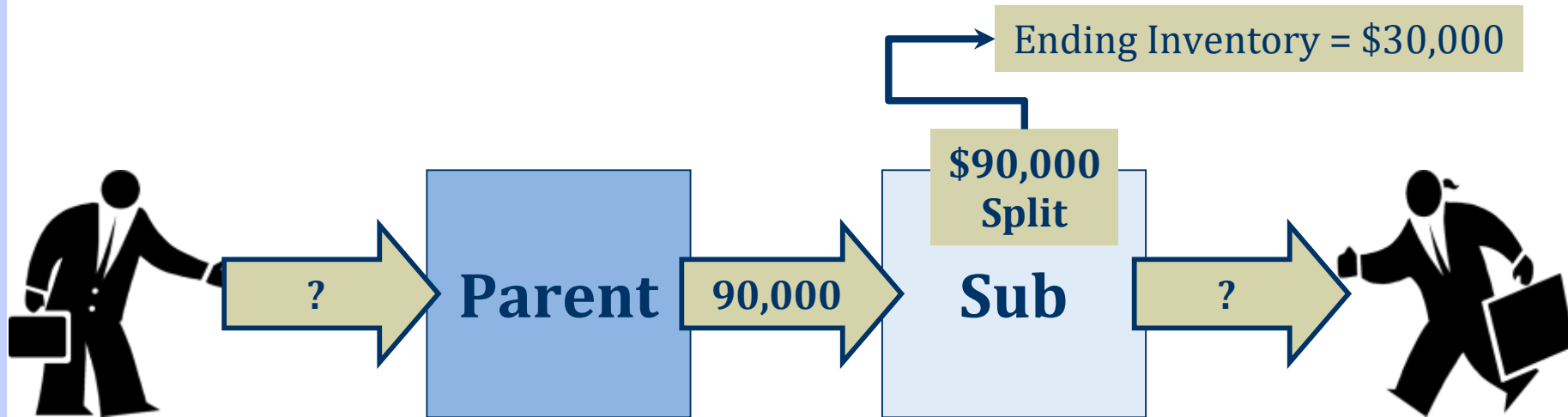
Practice Quiz Question #3

For 20X8, Post reported \$90,000 of intercompany sales (25% markup on cost and fully paid for by year end) to Script, which reported \$30,000 of intercompany acquired inventory at 12/31/X8. The unrealized profit at 12/31/X8 is

- a. \$0.
- b. \$6,000.
- c. \$7,500.
- d. \$30,000.
- e. None of the above.



Practice Quiz Question #3 Solution

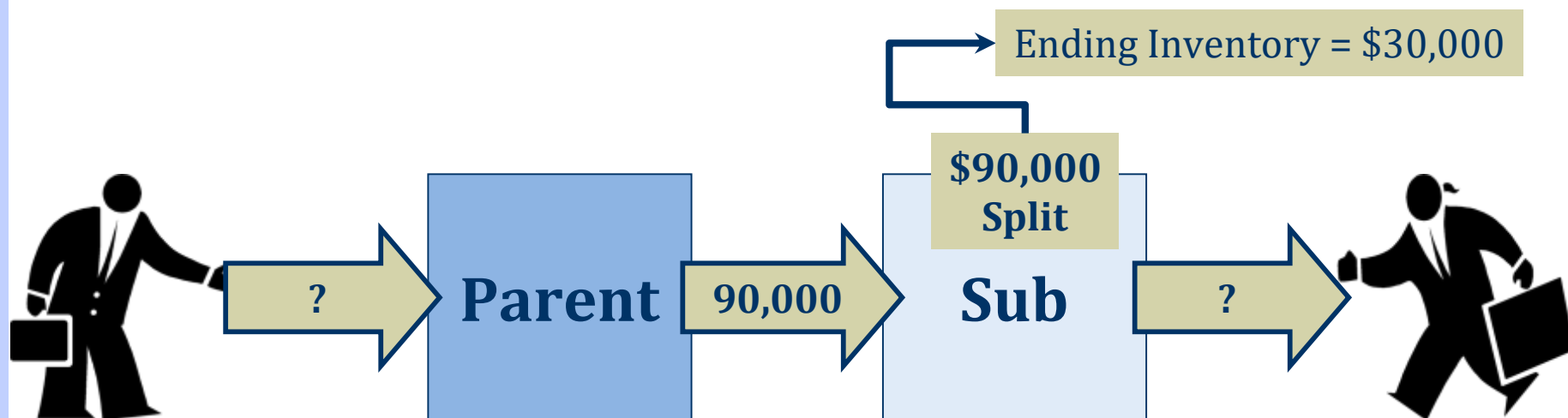


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	90,000		30,000
– COGS	C		
Gross Profit	0.25 C		?
Gross Profit %	?		



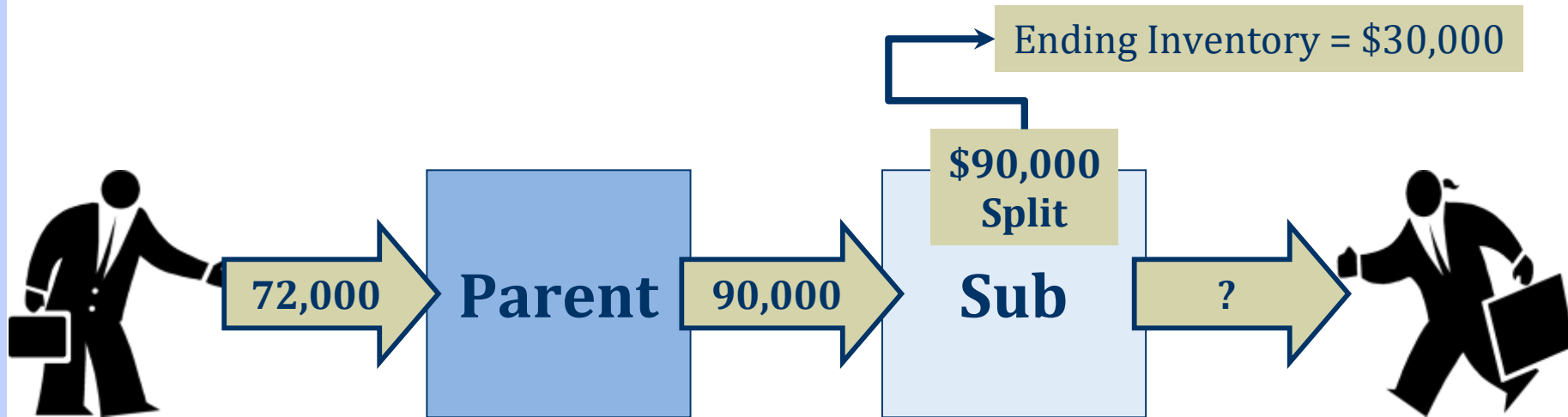
Practice Quiz Question #3 *Solution*



$$\begin{aligned} 90,000 - C &= 0.25 C \\ 1.25 C &= 90,000 \\ C &= 90,000 / 1.25 = 72,000 \end{aligned}$$



Practice Quiz Question #3 *Solution*

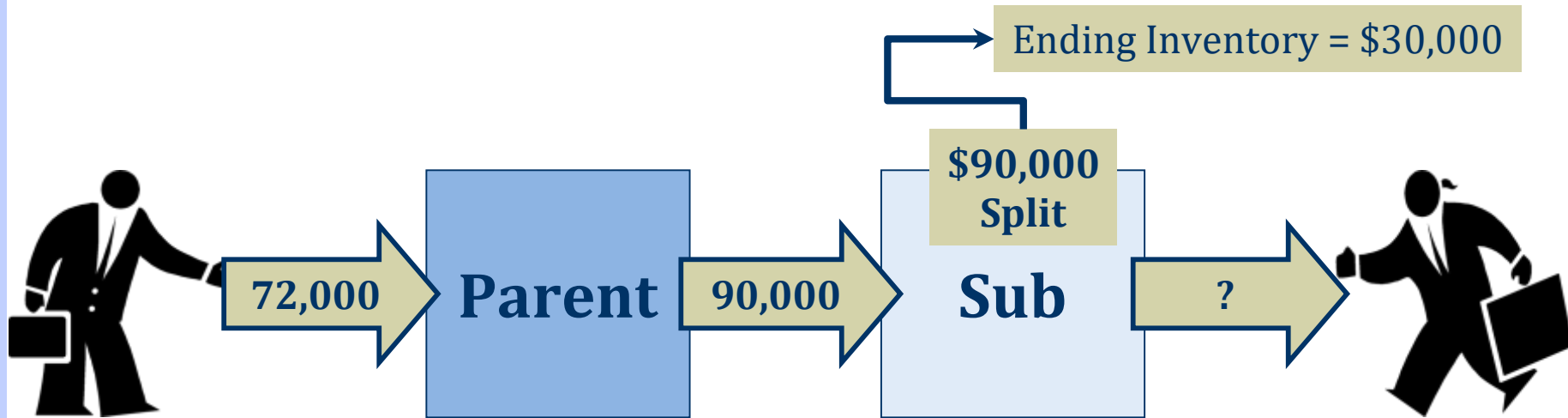


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	90,000		30,000
– COGS	72,000		
Gross Profit	18,000		?
Gross Profit %	20%		



Practice Quiz Question #3 Solution

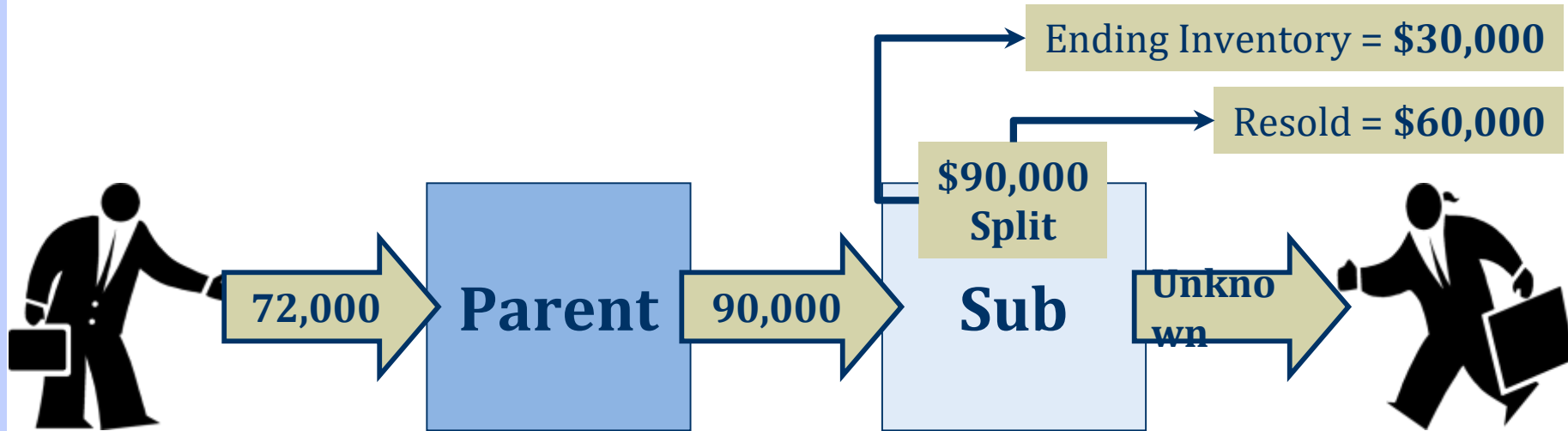


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	90,000		30,000
– COGS	72,000		
Gross Profit	18,000		6,000
Gross Profit %	20%		



Practice Quiz Question #3 Solution



What happened to it?

	Total Interco Sales	Resold	On hand
Sales	90,000	60,000	30,000
– COGS	72,000	48,000	24,000
Gross Profit	18,000	12,000	6,000
Gross Profit %	20%		



Practice Quiz Question #3 *Solution*

For 20X8, Post reported \$90,000 of intercompany sales (25% markup on cost and fully paid for by year end) to Script, which reported \$30,000 of intercompany acquired inventory at 12/31/X8. The unrealized profit at 12/31/X8 is

- a. \$0.
- b. \$6,000 ($\$30,000 \text{ EI} \times 0.20 \text{ GP\%}$).**
- c. \$7,500.
- d. \$30,000.
- e. None of the above.



Practice Quiz Question #4

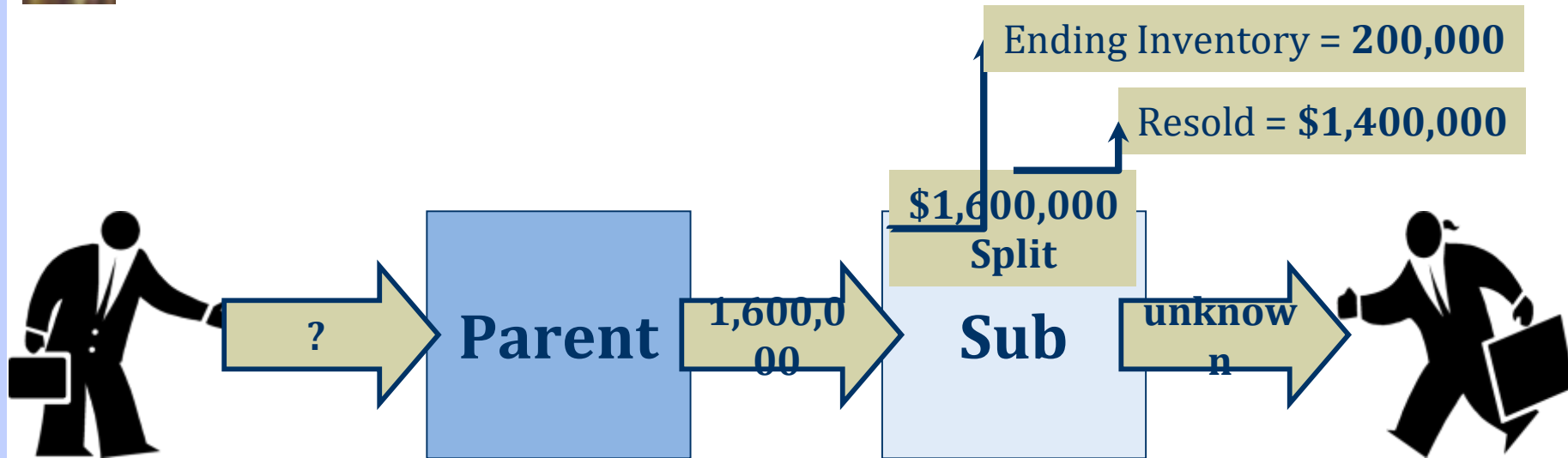
For 20X8, Sempre (80% owned by Para) reported \$1,600,000 of intercompany sales (1/3 markup on cost) to Para, which resold \$1,400,000 of this inventory by 12/31/X8.

The unrealized profit at 12/31/X8 is

- a. \$40,000.
- b. \$50,000.
- c. \$53,333.
- d. \$66,667.
- e. None of the above.



Practice Quiz Question #4 *Solution*

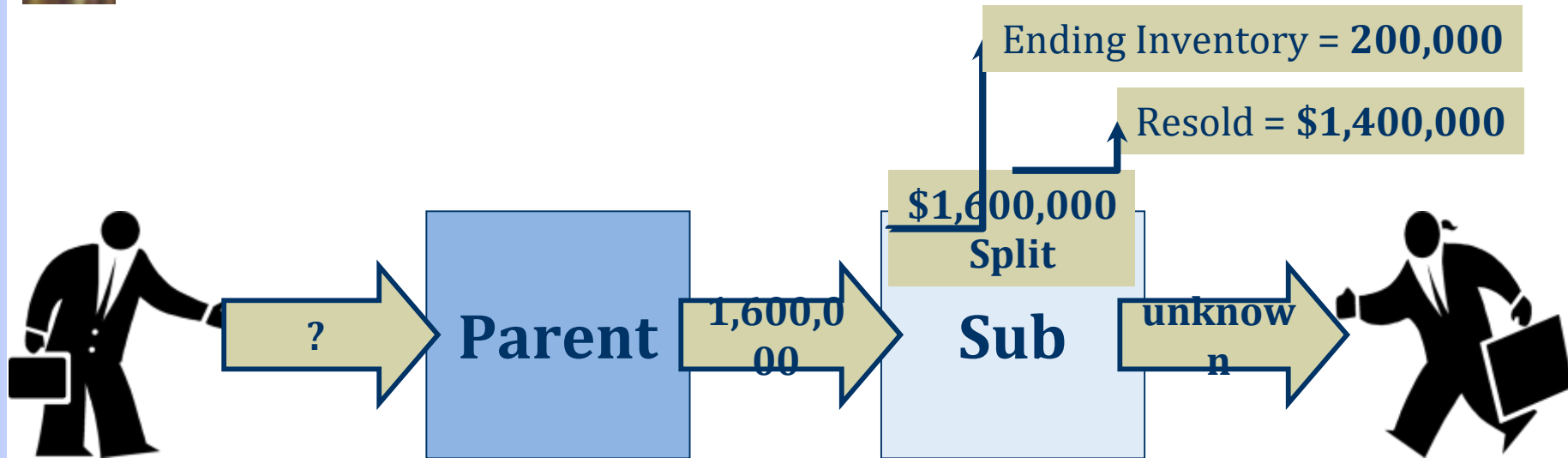


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,600,000	1,400,000	
– COGS			
Gross Profit			?
Gross Profit %	?		



Practice Quiz Question #4 *Solution*

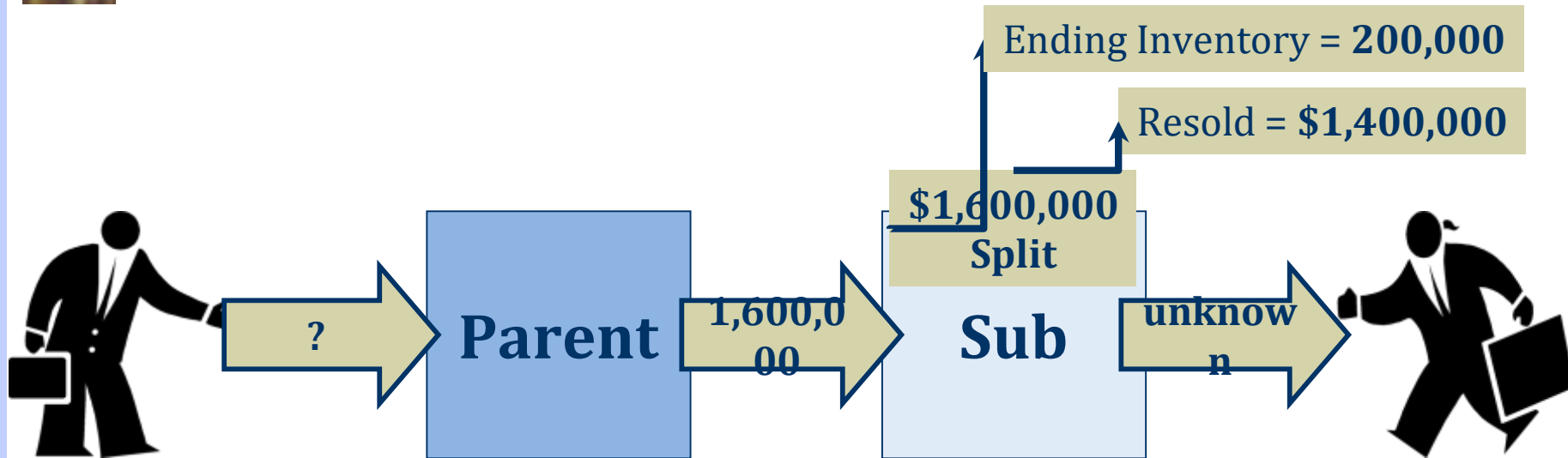


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,600,000	1,400,000	
– COGS	C		
Gross Profit	1/3 C		?
Gross Profit %	?		



Practice Quiz Question #4 *Solution*



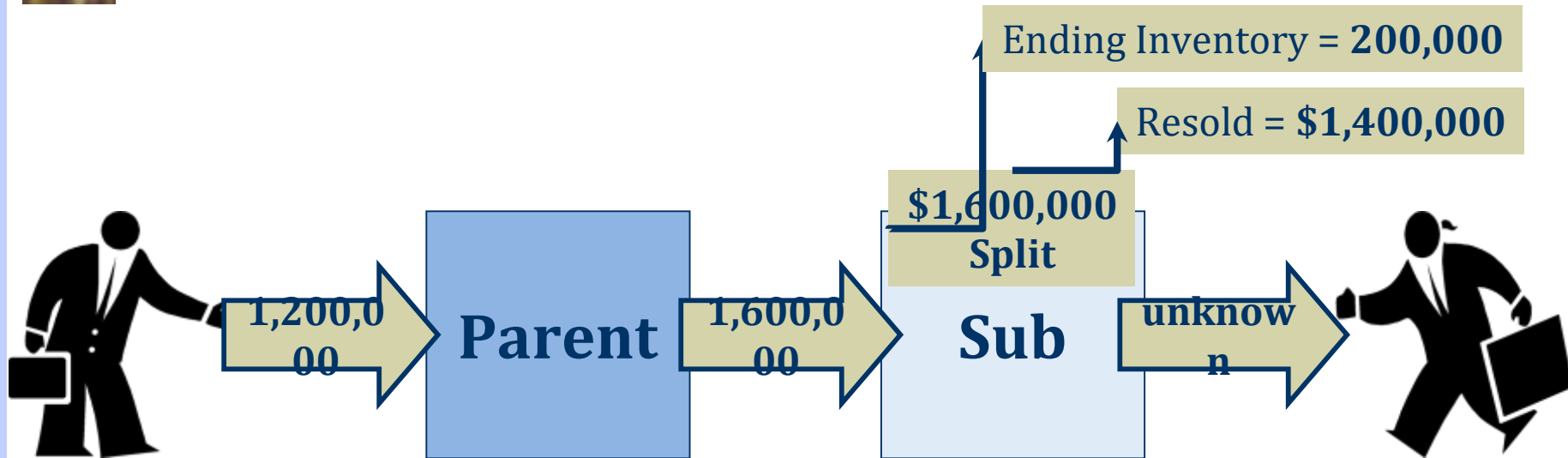
$$1,600,000 - C = \frac{1}{3} C$$

$$\frac{4}{3} C = 1,600,000$$

$$C = 1,600,000 / \left(\frac{4}{3}\right) = 1,200,000$$



Practice Quiz Question #4 *Solution*

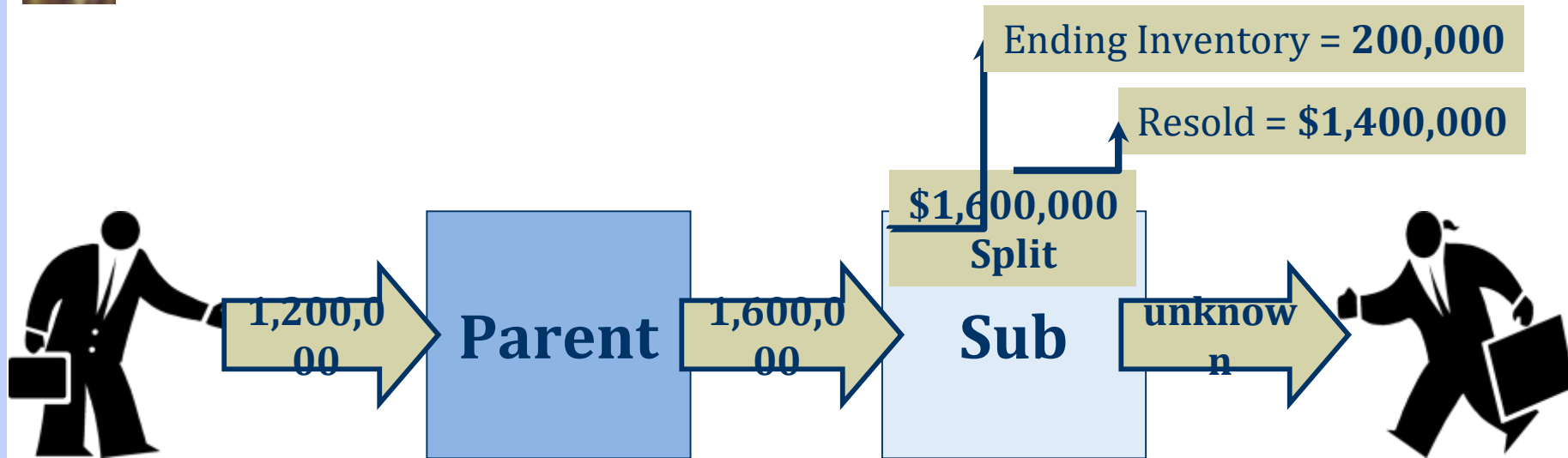


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,600,000	1,400,000	
– COGS	1,200,000		
Gross Profit	400,000		?
Gross Profit %	25%		



Practice Quiz Question #4 *Solution*

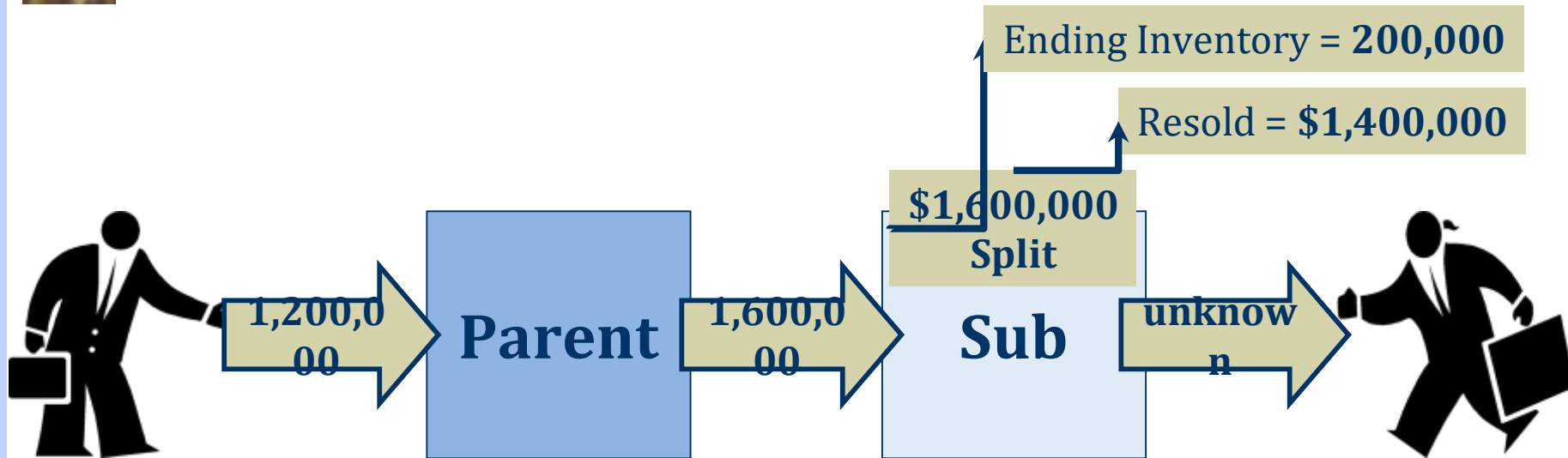


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,600,000	1,400,000	200,000
– COGS	1,200,000		
Gross Profit	400,000		?
Gross Profit %	25%		



Practice Quiz Question #4 *Solution*

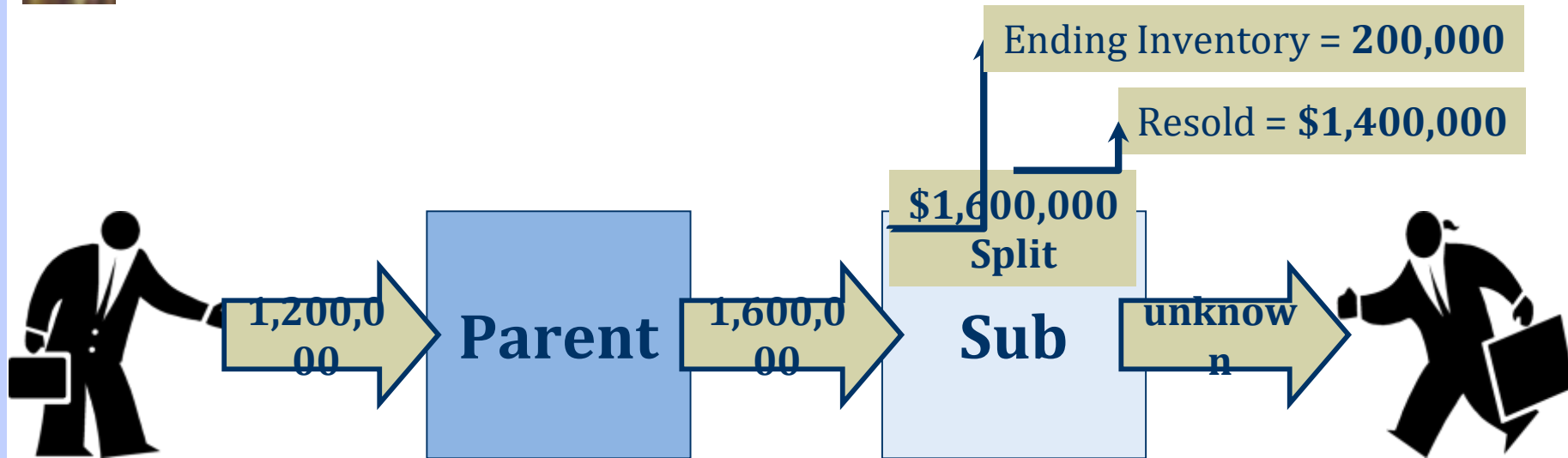


What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,600,000	1,400,000	200,000
– COGS	1,200,000		
Gross Profit	400,000		50,000
Gross Profit %	25%		



Practice Quiz Question #4 *Solution*



What happened to it?

	Total Interco Sales	Resold	On hand
Sales	1,600,000	1,400,000	200,000
– COGS	1,200,000	1,050,000	150,000
Gross Profit	400,000	350,000	50,000
Gross Profit %	25%		



Practice Quiz Question #4 *Solution*

For 20X8, Sempre (80% owned by Para) reported \$1,600,000 of intercompany sales (1/3 markup on cost) to Para, which resold \$1,400,000 of this inventory by 12/31/X8.

The unrealized profit at 12/31/X8 is

- a. \$40,000.
- b. \$50,000 ($\$200,000 \text{ EI} \times 0.25 \text{ GP\%}$).
- c. \$53,333.
- d. \$66,667.
- e. None of the above.



Learning Objective 6-3

Prepare equity-method journal entries and elimination entries for the consolidation of a subsidiary following downstream inventory transfers.



What to Look For

◆ Most problems will contain

- Inventory transferred from parent to sub (downstream),
or
- Inventory transferred from sub to parent (upstream).

◆ Often part of the inventory is sold to an outsider, but part remains in the buyer's ending inventory.

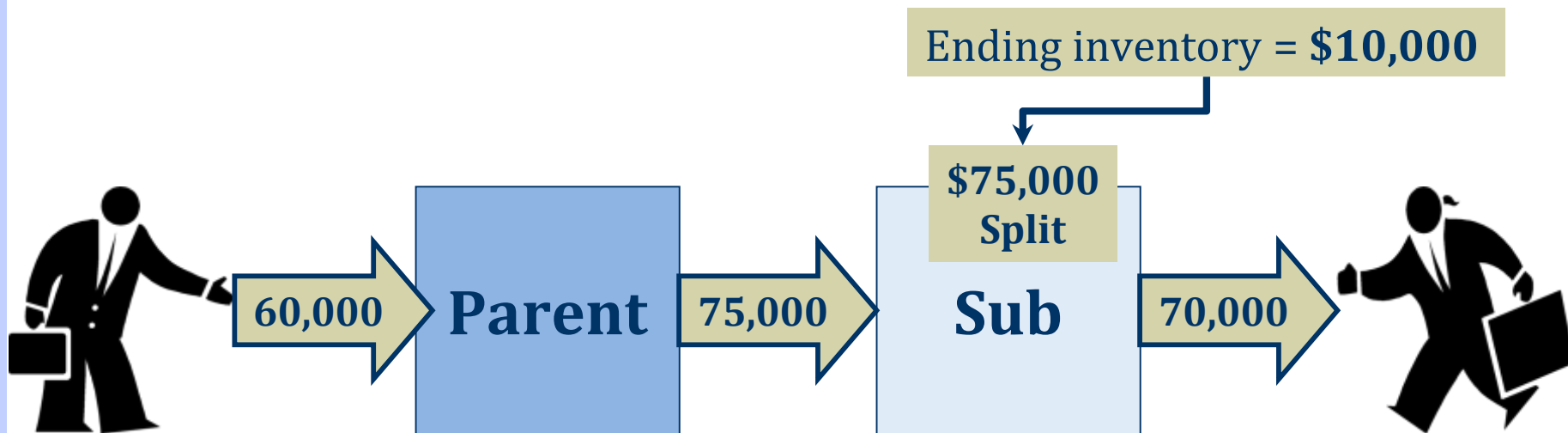
◆ Key: Any problem can be split into two parts

- The portion of the inventory that is sold
- The portion of the inventory that is still on hand



A Comprehensive Downstream Example

During 20X8, Parent sold inventory originally costing \$60,000 to its 100% owned Sub for \$75,000. Sub sold most of the inventory purchased from Parent (all but \$10,000) for \$70,000 to outsiders during the year.





A Comprehensive Downstream Example

During 20X8, Parent sold inventory originally costing \$60,000 to its 100% owned Sub for \$75,000. Sub sold most of the inventory purchased from Parent (all but \$10,000) for \$70,000 to outsiders during the year.

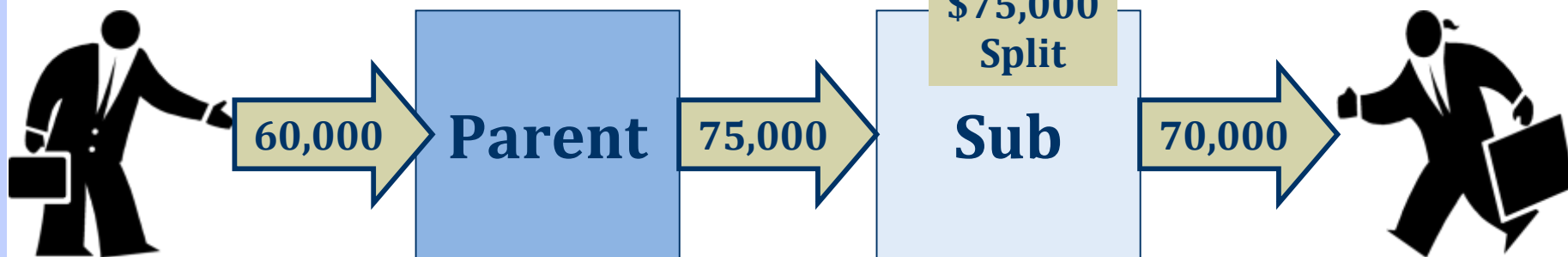
Income Statements

	Parent	Sub
Sales	\$75,000	\$70,000
Cost of sales	60,000	65,000
Gross profit	\$15,000	\$ 5,000

What happened to it?

Sold	On-hand
\$65,000	\$10,000
$20\% = \$2,000$	
Unrealized GP	

Ending inventory = \$10,000





One Approach: Split into Two Transactions

◆ This transaction can be broken into two pieces:

- Parent sells Sub inventory with a cost of \$52,000 for \$65,000. Sub then sells this inventory to outsiders for \$70,000.
- Parent sells Sub inventory with a cost of \$8,000 for \$10,000, which remains on hand in Sub's ending inventory.

	Total	Sold	On hand
Sales	\$75,000	\$65,000	\$10,000
– COGS	60,000	52,000	8,000
Gross Profit	\$15,000	\$13,000	\$ 2,000



Summary

To eliminate sale from Parent to Sub to Outsider :

Sales (Parent)	65,000	
Cost of Goods Sold (Sub)		65,000

To eliminate sale from Parent to Sub, not yet to Outsider:

Sales (Parent)	10,000	
Cost of Goods Sold (Parent)		
8,000		
Inventory (basis correction)		
2,000		

Can combine the two entries:

Sales	75,000	
Cost of Goods Sold		73,000
Inventory		2,000



Partial Consolidated Worksheet

	Parent	Sub	DR	CR	Consolidated
Income Statement					
Sales	75,000	70,000	75,000		70,000
COGS	60,000	65,000		73,000	52,000
Gross Profit	15,000	5,000	75,000	73,000	18,000
Balance Sheet					
Inventory	0	10,000		2,000	8,000



Second Approach: Short Cut Method

	Total	Sold	On hand
Sales	\$75,000	\$65,000	\$10,000
– COGS	60,000	52,000	8,000
Gross Profit	\$15,000	\$13,000	\$ 2,000

$$\text{COGS Credit} = \$65,000 + \$8,000$$

The numbers come right off the chart!

Sales	75,000	
Cost of Goods Sold		73,000
Inventory	2,000	



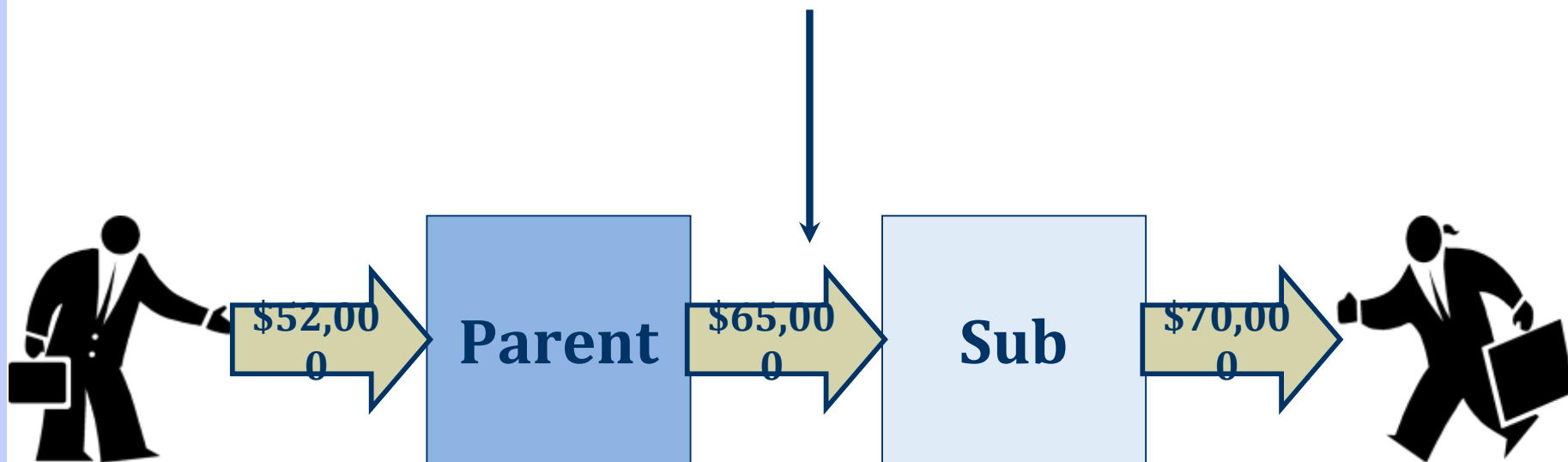
Part 1: Sale from Parent to Sub to Outsider

To eliminate sale from Parent to Sub to Outsider:

Sales (Parent) 65,000

Cost of Goods Sold (Sub) 65,000

Get rid of the non-arm's-length transaction!





Part 2: Sale from Parent to Sub (Not Outside)

Reverse the entire transaction!

To eliminate sale from Parent to Sub, not yet to Outsider:

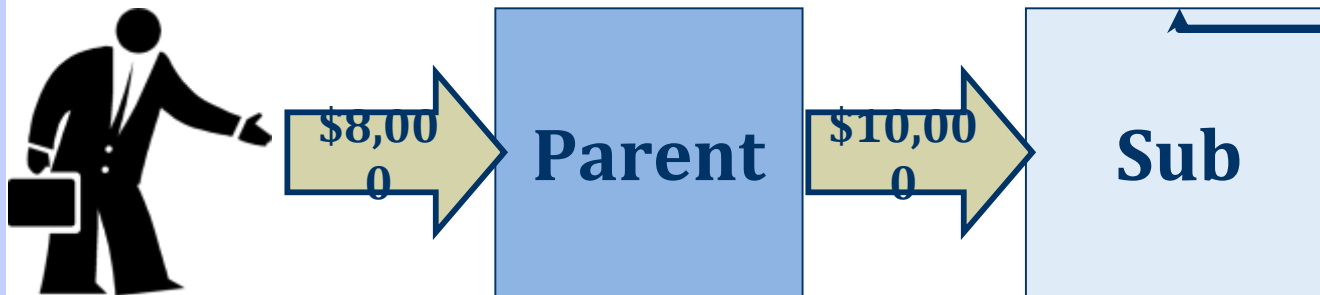
Sales (Parent)	10,000
Cost of Goods Sold (Parent)	
8,000	
Inventory (basis correction)	

2,000

Sales	\$10,000
Cost of sales	8,000
Gross profit	\$ 2,000

Parent's gross profit is overstated
by \$2,000

Sub's inventory is overstated by \$2,000





Partial Consolidated Worksheet

	Parent	Sub	DR	CR	Consol- idated
Income Statement					
Sales	75,000	70,000	75,000		70,000
COGS	60,000	65,000		73,000	52,000
Inc from Sub	5,000		5,000		
Net Income	20,000	5,000	80,000	73,000	18,000
Balance Sheet			Not the same!		
Inventory	0	10,000		2,000	8,000



Fully-adjusted Equity Method Adjustment

- ◆ Don't forget that one of the desirable properties of using the equity method is that the parent's net income should be equal to the consolidated net income.
- ◆ If you only adjust for unrealized deferred profit in the consolidation, the consolidated net income will be different from the parent's income!
 - ◆ Thus, an actual adjustment on the parent's books in addition to the worksheet entries above.
 - ◆ Like we did for the excess fair value amortization.



Fully-adjusted Equity Method Adjustment

- ◆ After calculating the unrealized deferred profit, simply make an extra adjustment to back it out.
- ◆ Do this at the same time you record the parent's share of the sub's income.

**Parent NI =
Consolidated NI**

Sales	\$75,000
COGS	60,000
Gross profit	<u>\$15,000</u>
Inc. from Sub	<u>3,000</u>
NI	\$18,000

Investment in Sub		Income from Sub	
NI 5,000	2,000	5,000 NI	2,000
	Unreal GP		3,000

Reverse next year when this inventory is sold!



Partial Consolidated Worksheet

	Parent	Sub	DR	CR	Consol -idated
Income Statement					
Sales	75,000	70,000	75,000		70,000
COGS	60,000	65,000		73,000	52,000
Inc from Sub	3,000		3,000		
Net Income	18,000	5,000	78,000	73,000	18,000
Balance Sheet			Now they're the same!		
Inventory	0	10,000		2,000	8,000



Practice Quiz Question #5

Under the fully adjusted equity method, what is one benefit of making an equity method adjustment to defer unrealized gross profit on inventory transfers?

- a. Consolidated net income always increases.
- b. Parent company net income always increases.
- c. Parent company net income is not equal to consolidated net income.
- d. Parent company net income equals consolidated net income.



Practice Quiz Question #5 *Solution*

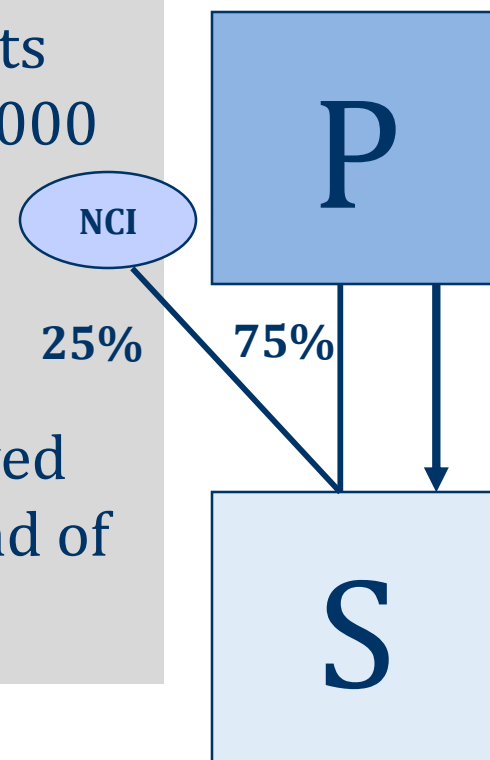
Under the fully adjusted equity method, what is one benefit of making an equity method adjustment to defer unrealized gross profit on inventory transfers?

- a. Consolidated net income always increases.
- b. Parent company net income always increases.
- c. Parent company net income is not equal to consolidated net income.
- d. Parent company net income equals consolidated net income.



Review Exercise Part 1: Downstream

- ◆ Para sold inventory costing \$100,000 to its **75%-owned** subsidiary, Shute, for \$125,000 in 20X8.
- ◆ Shute resold most of this inventory for \$230,000 in 20X8.
- ◆ At 12/31/X8, Shute's balance sheet showed intercompany-acquired inventory on hand of \$20,000.



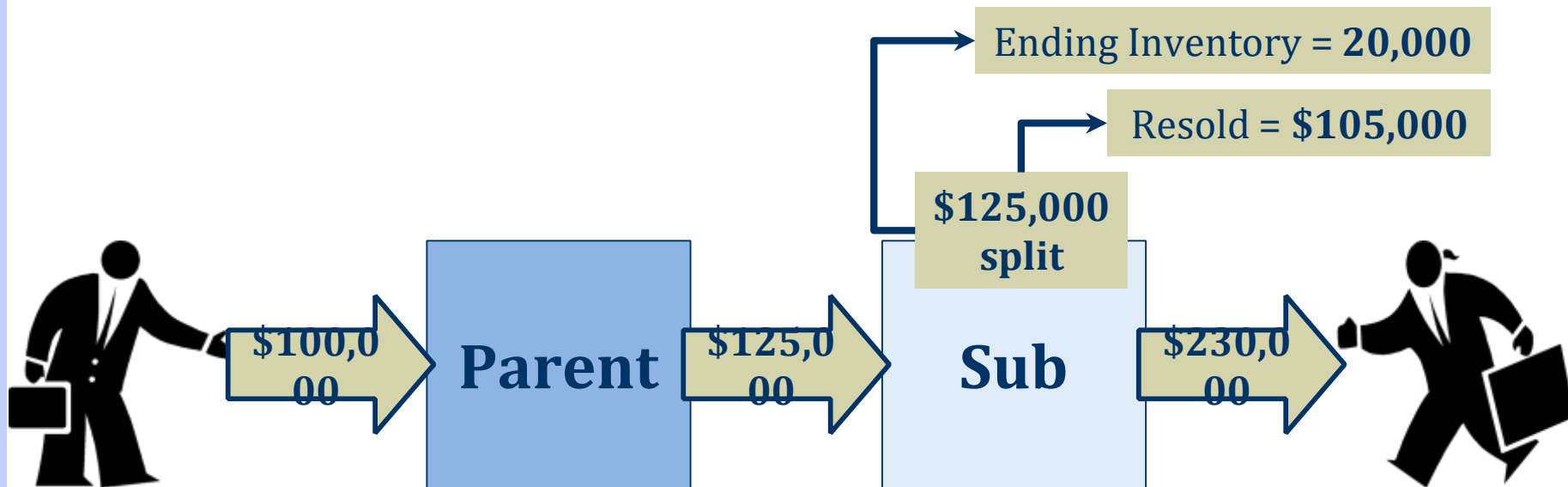
Required:

- ◆ Prepare the **consolidation entry** and/or entries required at 12/31/X8 under the **equity method**.
- ◆ **Since this is a DOWNSTREAM transaction, we don't share the GP deferral with the NCI.**



Review Exercise Part 1: Big Picture

	Total	Sold	On hand
Sales	125,000		20,000
– COGS	100,000		
Gross Profit	25,000		
Gross Profit %			

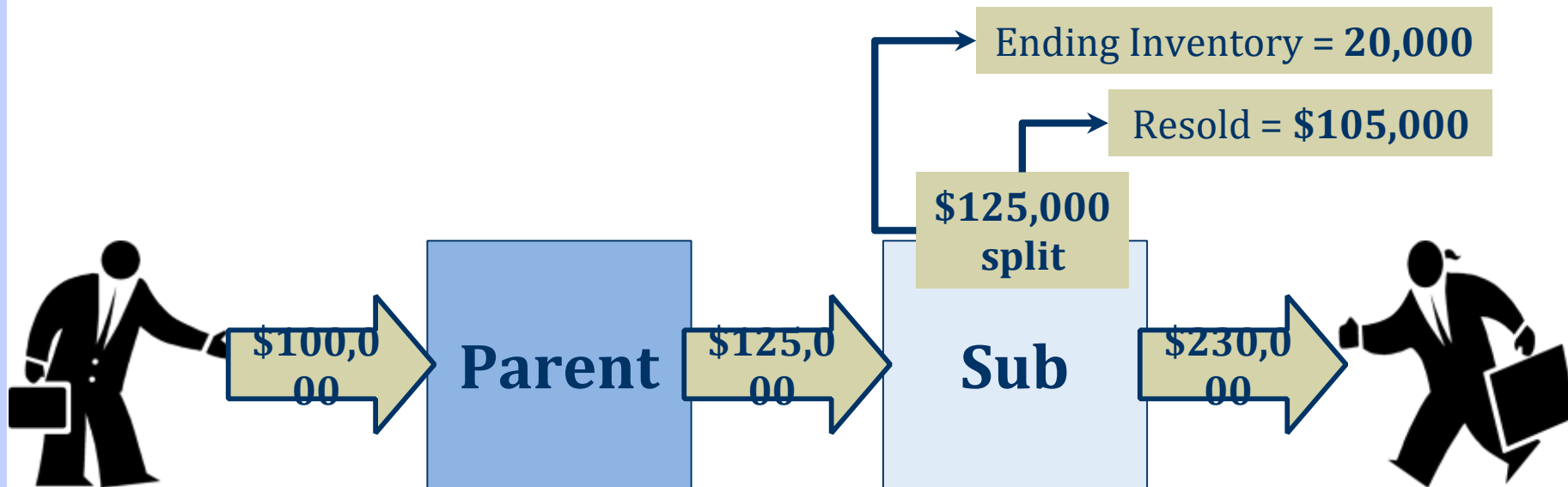




Review Exercise Part 1: Big Picture

	Total	Sold	On hand
Sales	125,000		20,000
– COGS	100,000		
Gross Profit	25,000		

$$\text{Gross Profit \%} = 25,000 / 125,000 = 1/5 = 20\%$$



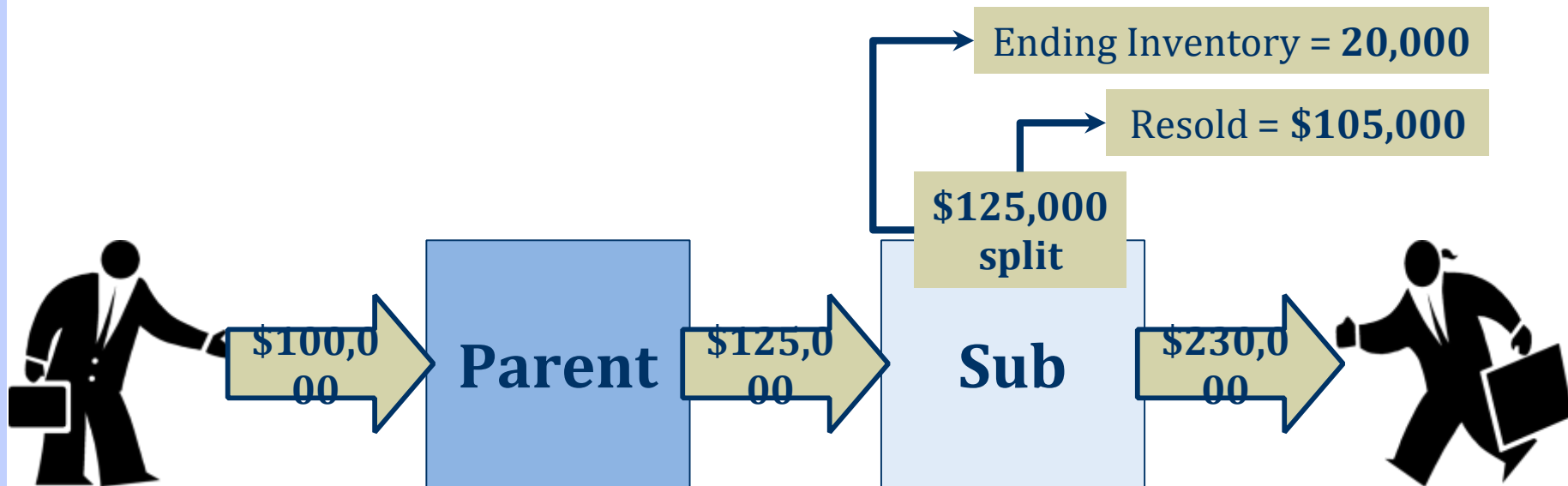


Review Exercise Part 1: Big Picture

	Total	Sold	On hand
Sales	125,000	105,000	20,000
– COGS	100,000	84,000	16,000
Gross Profit	25,000	21,000	4,000

Gross Profit % = $25,000 \div 125,000 = 1/5 = 20\%$

Unrealized GP



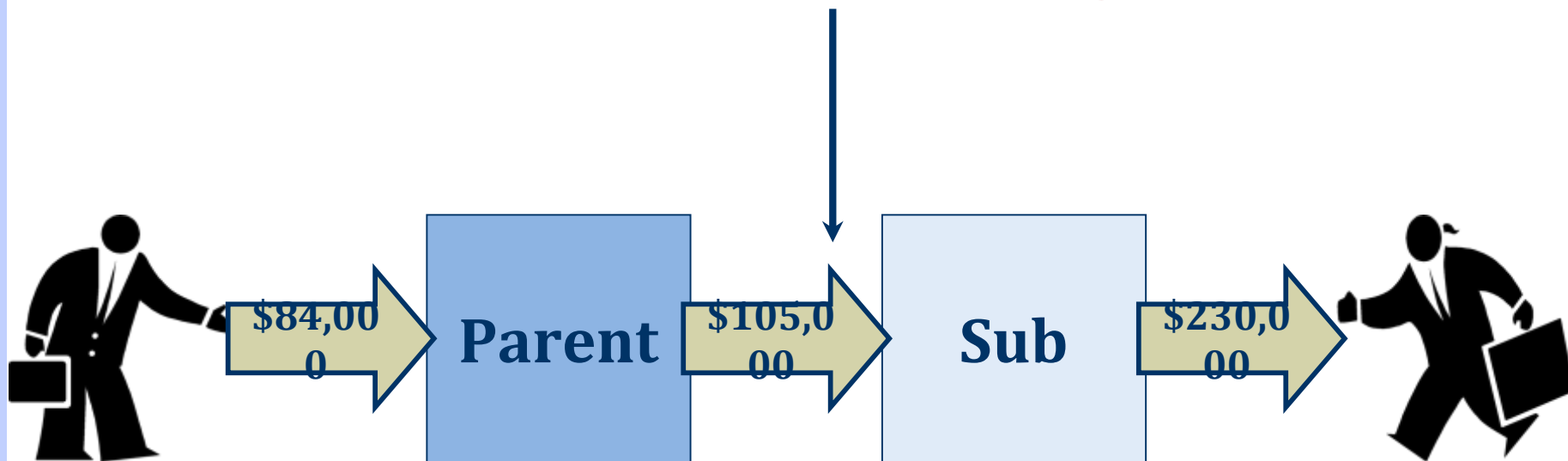


Review Exercise 1: Sale from Parent to Sub to Outsider

To eliminate sale from Parent to Sub to Outsider:

Sales (Parent)	105,000	
Cost of Goods Sold (Sub)		105,000

Get rid of the internal non-arm's-length transaction!





Review Exercise 1: Sale from Parent to Sub (Not Yet Outside)

Reverse the entire transaction!

To eliminate sale from Parent to Sub, not yet to Outsider:

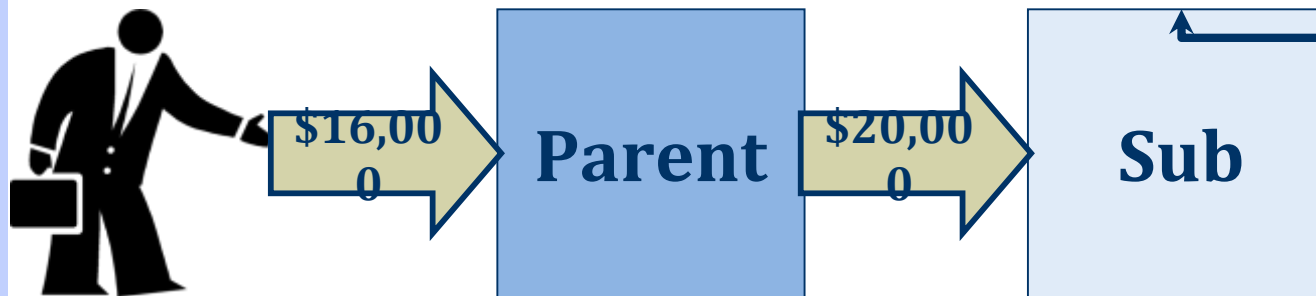
Sales (Parent)	20,000
Cost of Goods Sold (Parent)	
16,000	
Inventory (basis correction)	

4,000

Sales	\$20,000
Cost of sales	16,000
Gross profit	\$ 4,000

Parent's gross profit is overstated by \$4,000

Sub's inventory is overstated by \$4,000





Review Exercise 1: Summary

To eliminate sale from Parent to Sub to Outsider:

Sales (Parent)	105,000	
Cost of Goods Sold (Sub)		105,000

To eliminate sale from Parent to Sub, not yet to Outsider:

Sales (Parent)	20,000	
Cost of Goods Sold (Parent)		
16,000		
Inventory (basis correction)		

4,000

Combine both entries:

Sales	125,000	
Cost of Goods Sold		121,000
Inventory		4,000

Fully-adjusted Equity Method Entry on Parent's books:

Income from Sub	4,000	
Investment in Sub		4,000



Review Exercise Part 1: Short Cut

	Total	Sold	On hand
Sales	125,000	105,000	20,000
– COGS	100,000	84,000	16,000
Gross Profit	25,000	21,000	4,000

COGS Credit = 105,000 + 16,000 = 121,000

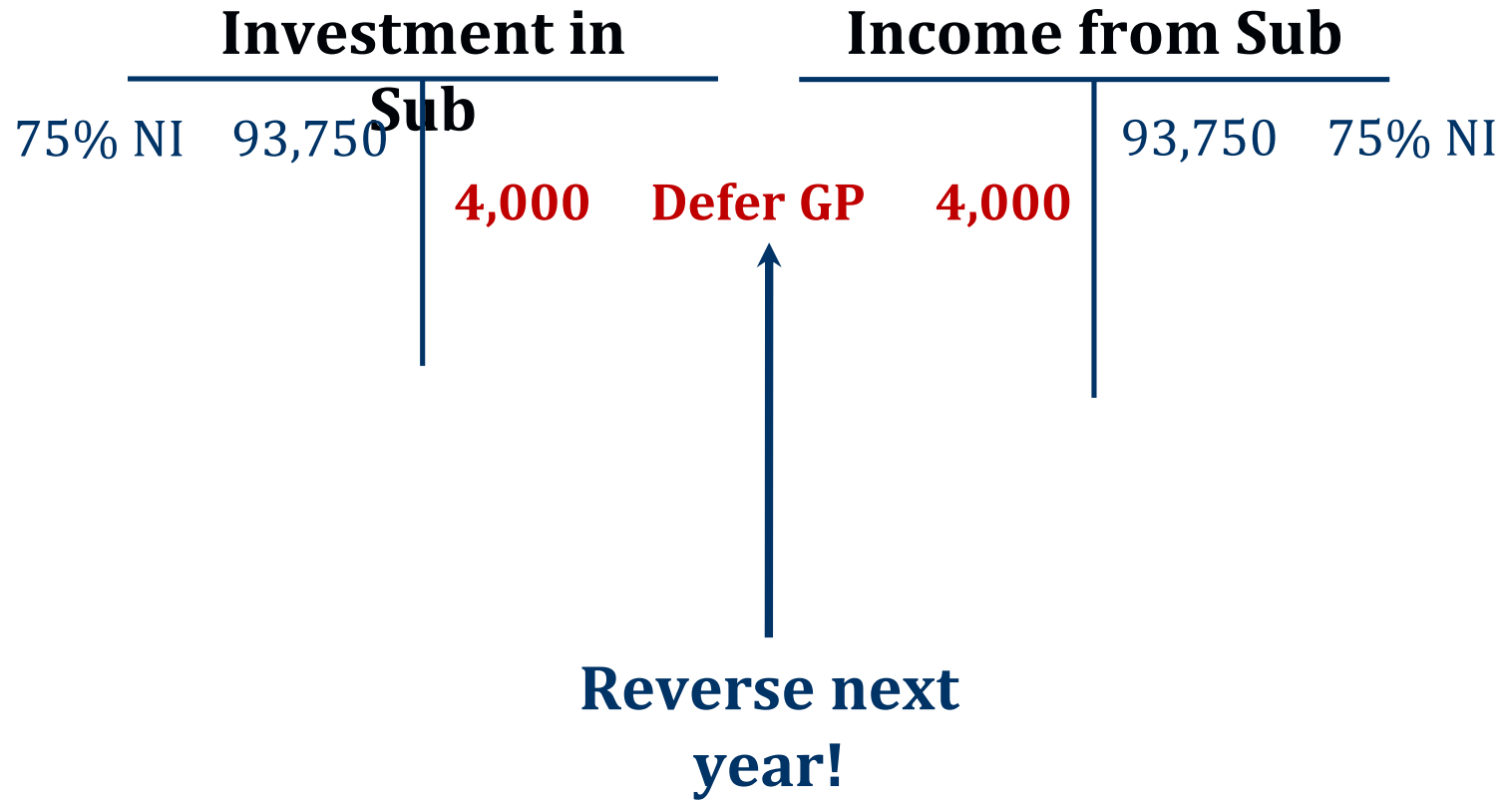
Unrealized GP

Worksheet Elimination Entry:

Sales	125,000	
Cost of Goods Sold		121,000
Inventory	4,000	



Review Exercise 1: Equity Method Entry





Review Exercise 1: Equity Method Reversal Next Year

Equity Method Adjustment on Parent's books in 20X7:

Income from Sub	4,000	
Investment in Sub		4,000

Reversal of 20X7 Deferral on Parent's books in 20X8:

Investment in Sub	4,000	
Income from Sub		4,000



Review Exercise Part 1

Worksheet Elimination Entry in Year 1:

Sales	125,000		
Cost of Goods Sold		121,000	
Inventory		4,000	

FYI, this year's deferral is REVERSED next year to recognize when sold!



Review Exercise 1: Equity Method Entry

Investment in			Income from Sub		
Sub					
75% NI	93,750			93,750	75% NI
		4,000	Defer GP	4,000	
Low	4,000			89,750	

Downstream, so don't split
the deferral with the NCI.



Review Exercise Part 1

Worksheet Elimination Entry in Year 1:

Sales	125,000	
Cost of Goods Sold		121,000
Inventory	4,000	

FYI, this year's deferral is REVERSED next year to recognize when sold!

Worksheet Elimination Entry in Year 2:

Investment in Sub	4,000	
Cost of Goods Sold		4,000

INCREASES income!



Review Exercise 1: Partial Consolidated Worksheet

	Parent	Sub	DR	CR	Consolidated
Income Statement					
Sales	125,000	230,000	125,000		230,000
COGS	100,000	105,000		121,000	84,000
Inc from Sub	89,750		89,750	Basic	
Gross Profit	114,750	125,000	214,750	121,000	146,000
NCI in NI			31,250	Basic	(31,250)
CI in NI	114,750	125,000	246,000	121,000	114,750
Balance Sheet					
Inventory		20,000		4,000	16,000



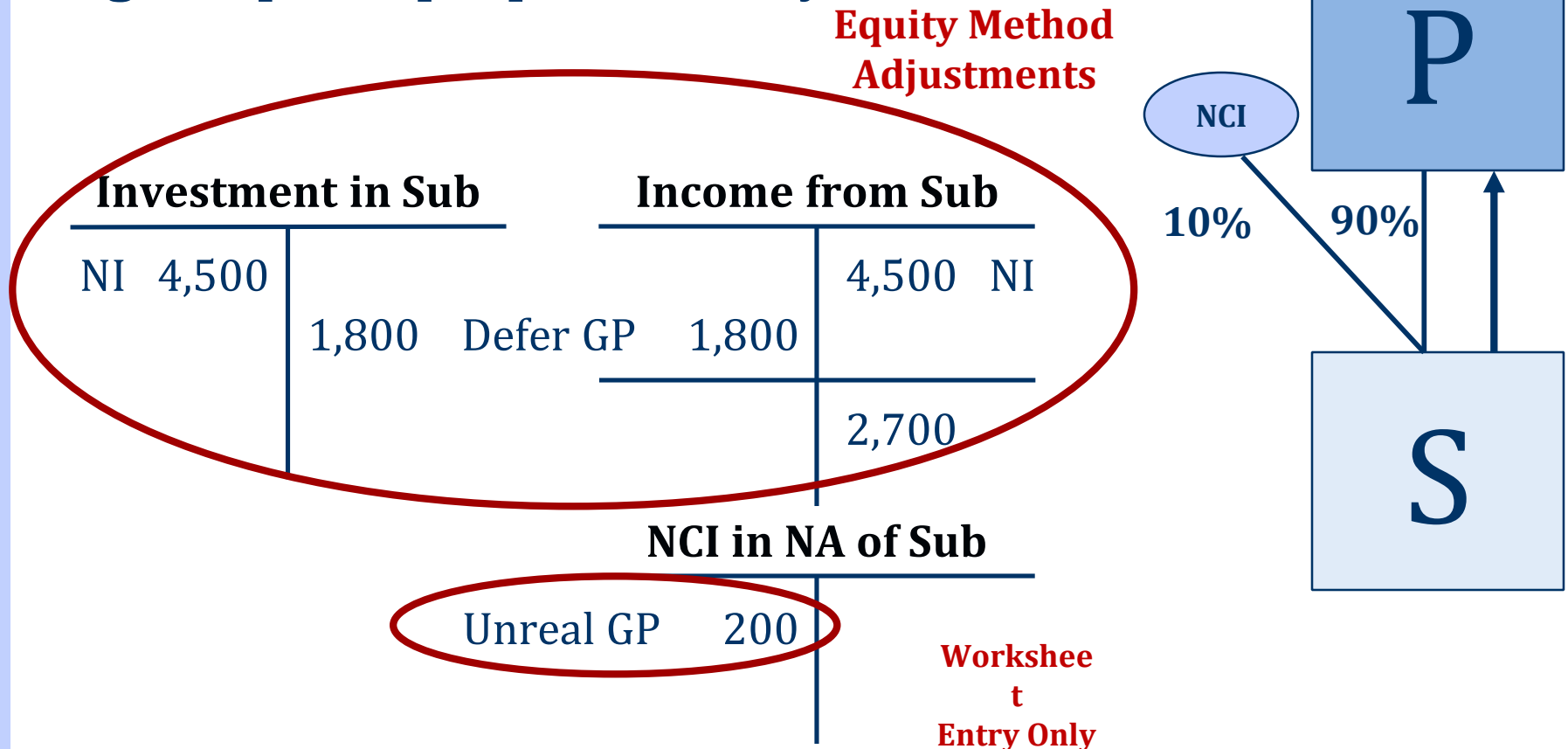
Learning Objective 6-4

Prepare equity-method journal entries and elimination entries for the consolidation of a subsidiary following upstream inventory transfers.



Partially Owned Upstream Sales

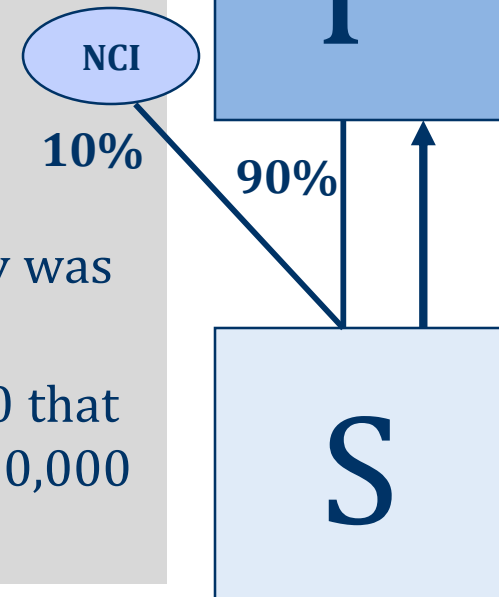
- ◆ Must share deferral with the NCI shareholders.
- ◆ Simply split up the adjustment for unrealized gross profit proportionately.





Review Exercise Part 2

- ◆ In 20X7, Sensei, a **90%-owned** subsidiary of Padawan, sold inventory to Padawan for \$600,000, which includes a markup of 25% on Sensei's cost.
- ◆ Padawan resold most of this inventory in 20X7 for \$588,000.
- ◆ At 12/31/X7, Padawan reported \$110,000 of this inventory in its balance sheet. (This ending inventory was resold in 20X8 by Padawan.)
- ◆ In 20X8, Sensei sold Padawan inventory for \$900,000 that had a cost of \$675,000, of which Padawan resold \$700,000 by 12/31/X8 for \$840,000.



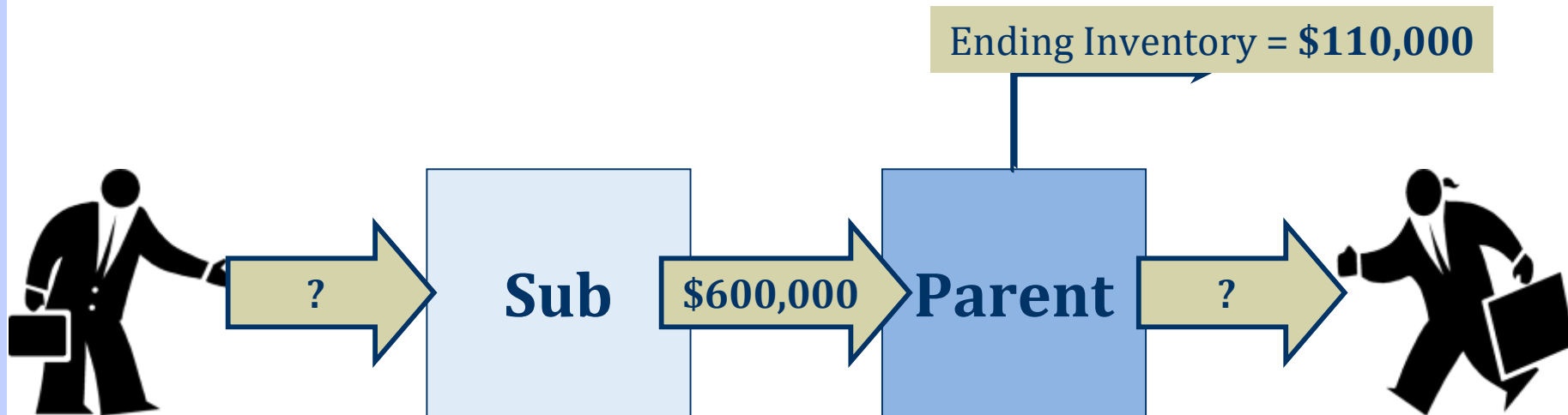
Required:

- ◆ Prepare the **consolidation entry** and/or entries required at 12/31/X8 under the **equity method**.
- ◆ Since this is an **UPSTREAM** transaction, we **do** share the GP deferral with the NCI.



Review Exercise Part 2: The Big Picture—20X7

	Total	Sold	On hand
Sales	600,000		110,000
– COGS	C		
Gross Profit	0.25C		



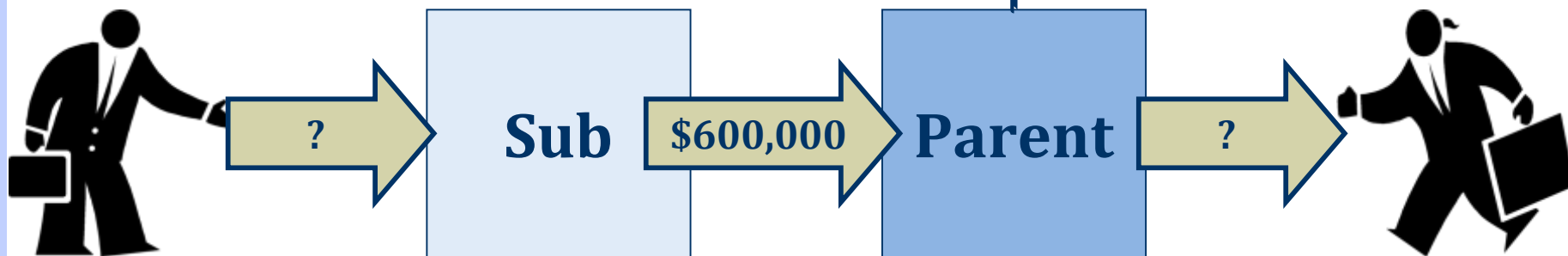


Review Exercise Part 2: The Big Picture—20X7

	Total	Sold	On hand
Sales	600,000		110,000
– COGS			
Gross Profit			

$$\begin{aligned} \$600,000 - C &= 0.25C \\ C &= \\ \$600,000 / 1.25 &= \\ \$480,000 \end{aligned}$$

Ending Inventory = \$110,000





Review Exercise Part 2: The Big Picture—20X7

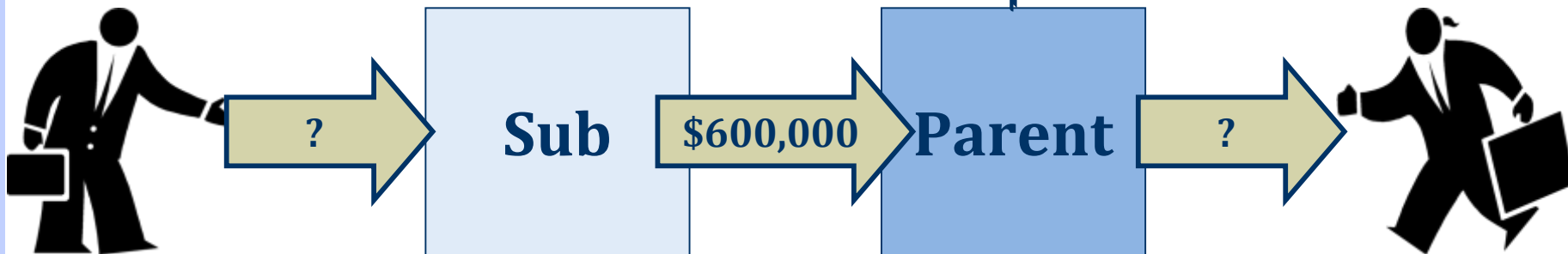
	Total	Sold	On hand
Sales	600,000		110,000
– COGS	480,000		
Gross Profit	120,000		

Gross Profit % = $120,000 / 600,000 = 1/5 = 20\%$

Unrealized GP

$$\begin{aligned} \$600,000 - C &= 0.25C \\ C &= \\ \$600,000 / 1.25 &= \\ \$480,000 \end{aligned}$$

Ending Inventory = \$110,000





Review Exercise Part 2: The Big Picture—20X7

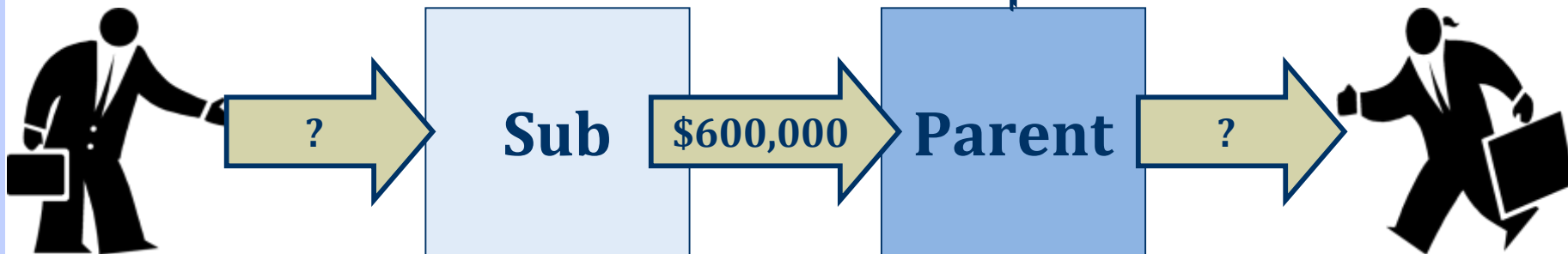
	Total	Sold	On hand
Sales	600,000		110,000
– COGS	480,000		
Gross Profit	120,000		22,000

Gross Profit % = $120,000 / 600,000 = 1/5 = 20\%$

Unrealized GP

$$\begin{aligned} \$600,000 - C &= 0.25C \\ C &= \\ \$600,000 / 1.25 &= \\ \$480,000 \end{aligned}$$

Ending Inventory = \$110,000





Review Exercise Part 2: The Big Picture—20X7

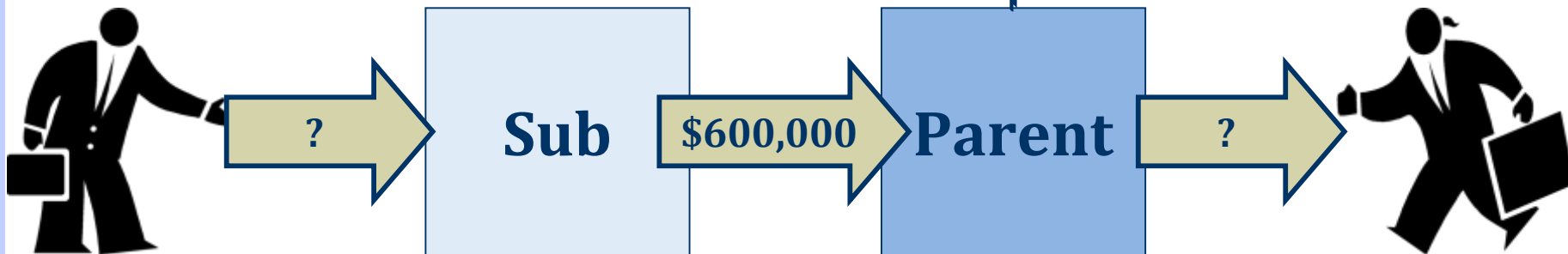
	Total	Sold	On hand
Sales	600,000	490,000	110,000
– COGS	480,000	392,000	88,000
Gross Profit	120,000	98,000	22,000

Gross Profit % = $120,000 / 600,000 = 1/5 = 20\%$

Unrealized GP

$$\begin{aligned} \$600,000 - C &= 0.25C \\ C &= \\ \$600,000 / 1.25 &= \\ \$480,000 \end{aligned}$$

Ending Inventory = \$110,000



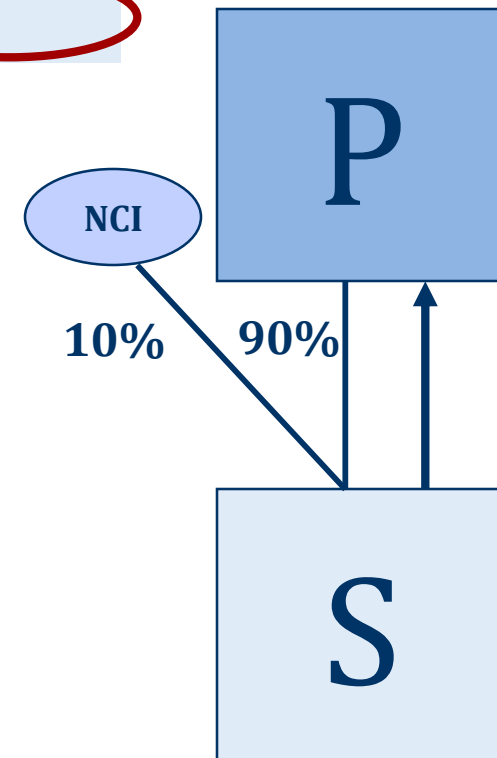


20X7 Upstream Sales: Elimination Entries— 20X7 & 20X8

20X7 Worksheet Elimination Entry:

Sales	600,000	
Cost of Goods Sold		578,000
Inventory	22,000	

Deferred GP this year “reversed”
to recognize in the financial
statements next year when sold.





20X7 Upstream Sales: Equity Method Adjustments — 20X7 & 20X8

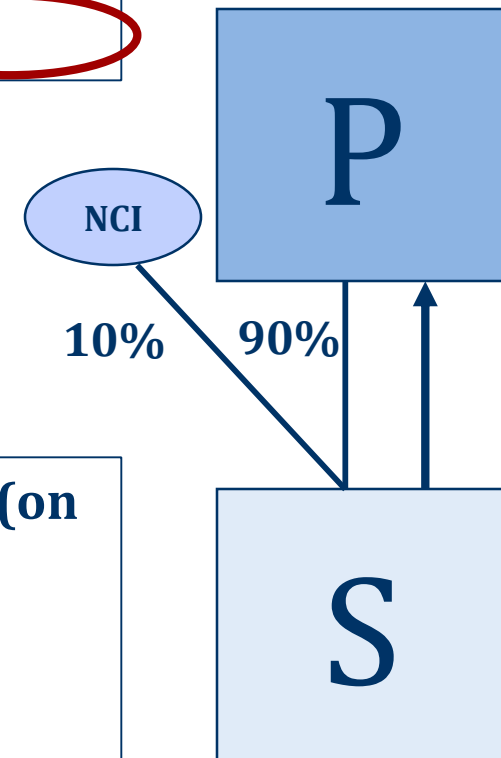
20X7 Equity Method Adjustment on Parent's books:

Income from Sub	19,800
Investment in Sub	19,800

Deferral of GP in 20X7
because not yet sold this year.

20X8 Equity Method Reversal of 20X7 Deferral (on Parent's books):

Investment in Sub	19,800
Income from Sub	19,800





20X7 Upstream Sales: 20X7 Equity Accounts

Investment in		Income from Sub	
Sub			
90% NI	108,000		108,000 90% NI
	19,800	X7 Deferral	19,800
Low	19,800		88,200



20X7 Upstream Sales: Elimination Entries— 20X7 & 20X8

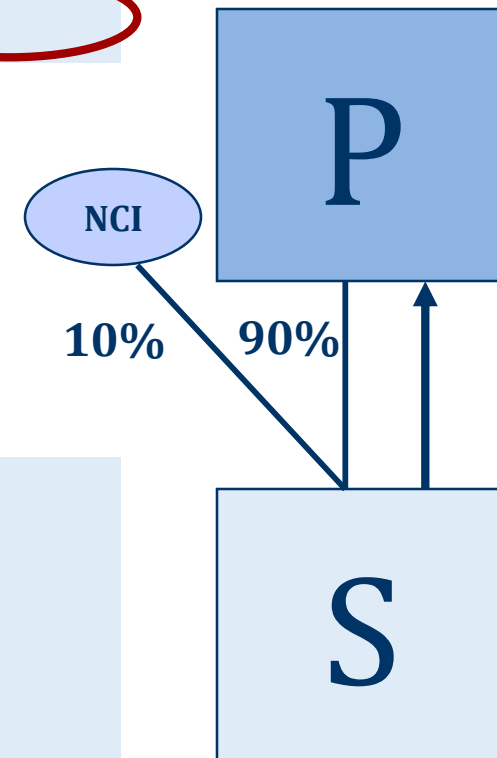
20X7 Worksheet Elimination Entry:

Sales	600,000	
Cost of Goods Sold		578,000
Inventory	22,000	

Deferred GP this year “reversed”
to recognize in the financial
statements next year when sold.

20X8 Worksheet Elimination Entry:

Investment in Sub	19,800	
NCI in NA of Sub	2,200	
Cost of Goods Sold		22,000





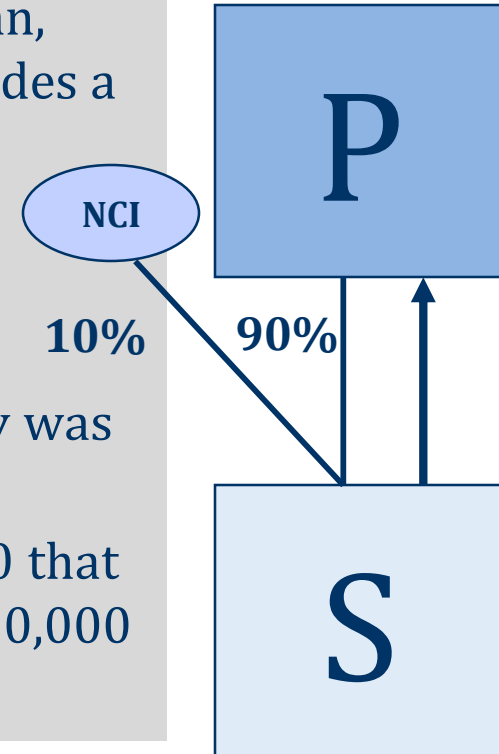
20X7 Upstream Sales: 20X7 Partial Worksheet

	Parent	Sub	DR	CR	Consolidated
Income Statement					
Sales	588,000	600,000	600,000		588,000
COGS	490,000	480,000		578,000	392,000
Inc from Sub	88,200		88,200	Basic	
Gross Profit	186,200	120,000	688,200	578,000	196,000
NCI in NI			9,800	Basic	(9,800)
CI in NI	186,200	120,000	698,000	578,000	186,200
Balance Sheet					
Inventory	110,000			22,000	88,000



Review Exercise Part 2

- ◆ In 20X7, Sensei, a **90%-owned** subsidiary of Padawan, sold inventory to Padawan for \$600,000, which includes a markup of 25% on Sensei's cost.
- ◆ Padawan resold most of this inventory in 20X7 for \$588,000.
- ◆ At 12/31/X7, Padawan reported \$110,000 of this inventory in its balance sheet. (This ending inventory was resold in 20X8 by Padawan.)
- ◆ In 20X8, Sensei sold Padawan inventory for \$900,000 that had a cost of \$675,000, of which Padawan resold \$700,000 by 12/31/X8 for \$840,000.



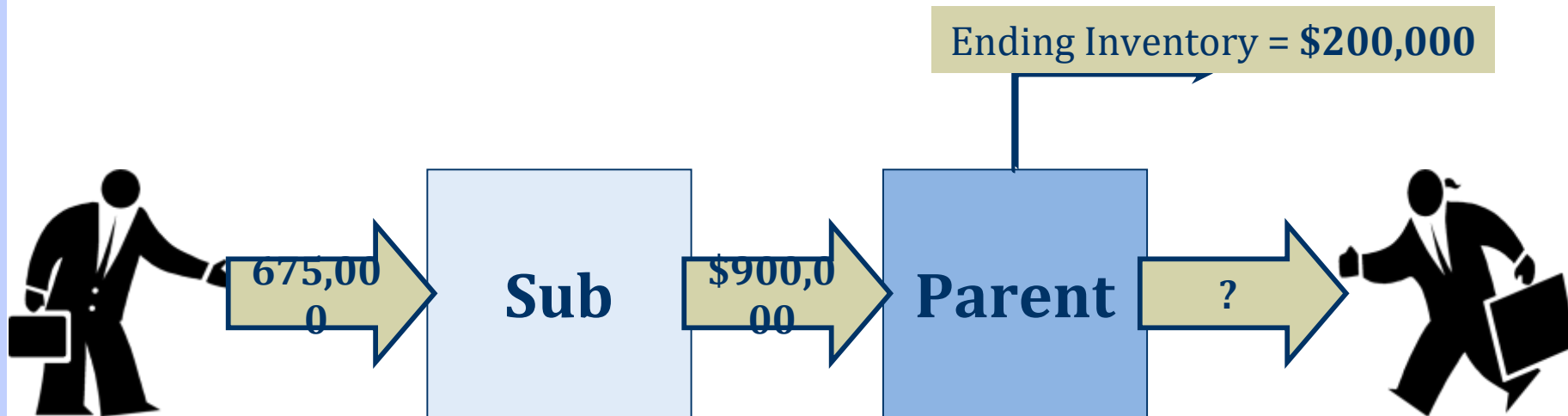
Required:

- ◆ Prepare the **consolidation entry** and/or entries required at 12/31/X8 under the **equity method**.
- ◆ Since this is an **UPSTREAM** transaction, we **do** share the GP deferral with the NCI.



Review Exercise Part 2: The Big Picture—20X8

	Total	Sold	On hand
Sales	900,000	700,000	
– COGS	675,000		
Gross Profit			

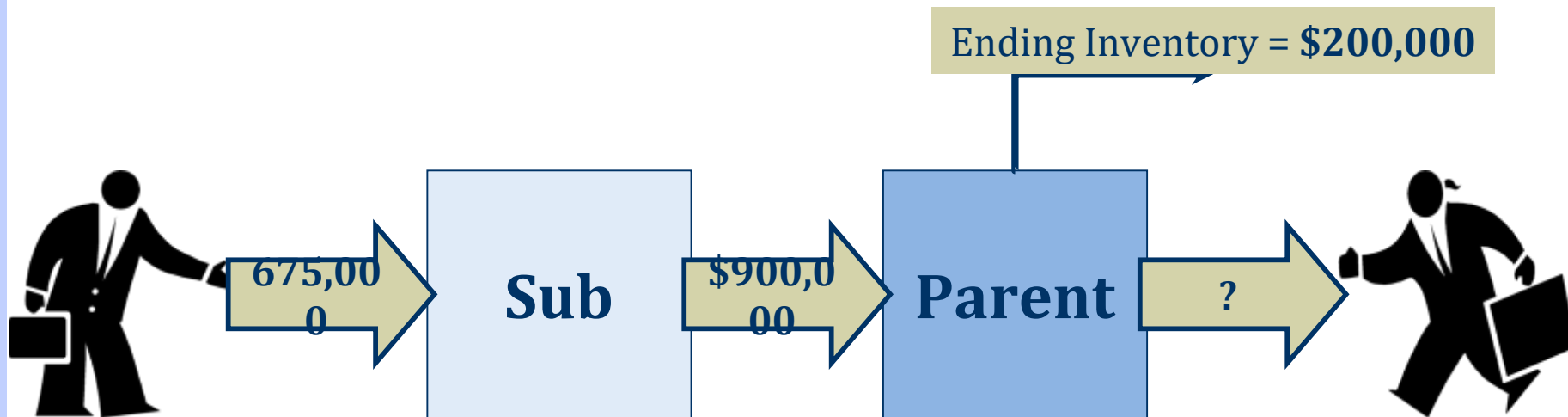




Review Exercise Part 2: The Big Picture—20X8

	Total	Sold	On hand
Sales	900,000	700,000	200,000
– COGS	675,000		
Gross Profit	225,000		

Gross Profit % = $225,000 / 900,000 = 0.25$



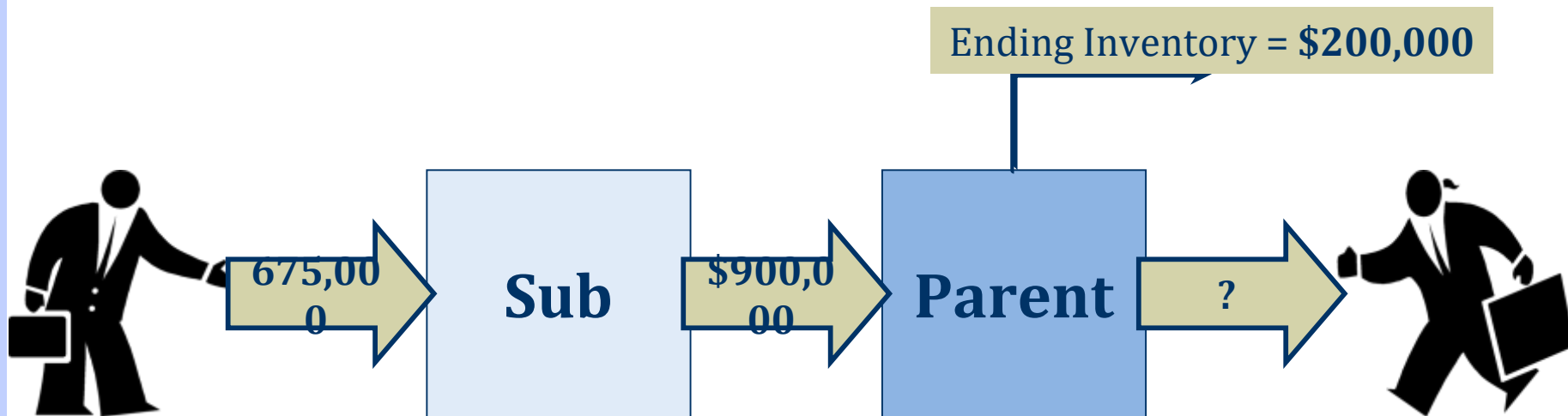


Review Exercise Part 2: The Big Picture—20X8

	Total	Sold	On hand
Sales	900,000	700,000	200,000
– COGS	675,000		
Gross Profit	225,000		

Gross Profit % = $225,000 / 900,000 = 0.25$

Unrealized GP



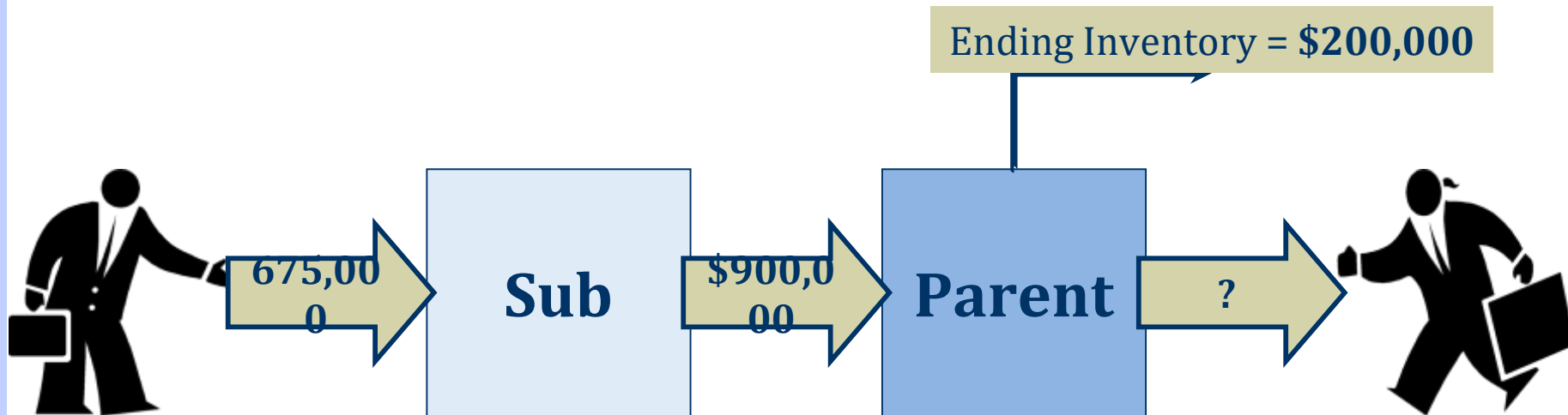


Review Exercise Part 2: The Big Picture—20X8

	Total	Sold	On hand
Sales	900,000	700,000	200,000
– COGS	675,000		
Gross Profit	225,000		50,000

Gross Profit % = $225,000 / 900,000 = 0.25$

Unrealized GP

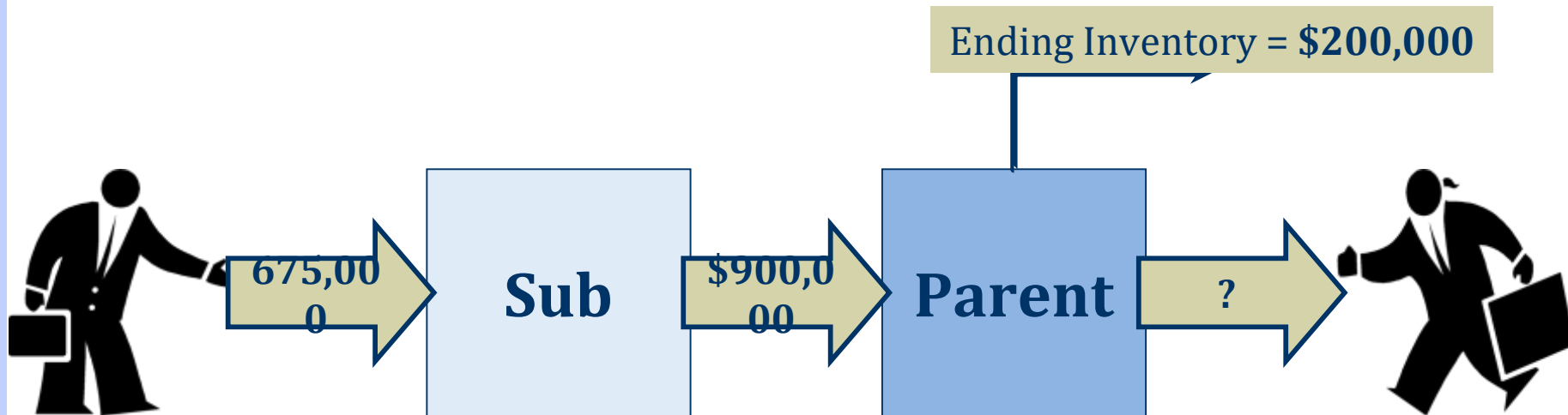




Review Exercise Part 2: The Big Picture—20X8

	Total	Sold	On hand
Sales	900,000	700,000	200,000
– COGS	675,000	525,000	150,000
Gross Profit	225,000	175,000	50,000
Gross Profit % = 225,000 / 900,000 = 0.25			

Unrealized GP





Review Exercise 2: Summary

To eliminate sale from Sub to Parent to Outsider:

Sales (Sub)	700,000
Cost of Goods Sold (Parent)	
700,000	

To eliminate sale from Sub to Parent, not yet to Outsider:

Sales (Sub)	200,000
Cost of Goods Sold (Sub)	150,000
Inventory (basis correction)	
50,000	

Combine both entries:

Sales	900,000
Cost of Goods Sold	850,000
Inventory	50,000

Fully-adjusted Equity Method Entry on Parent's books:

Income from Sub	45,000
Investment in Sub	45,000



Review Exercise 2: Short Cut

	Total	Sold	On hand
Sales	900,000	700,000	200,000
– COGS	675,000	525,000	150,000
Gross Profit	225,000	175,000	50,000

$$\text{COGS CR} = 700,000 + 150,000 = 850,000$$

The Elimination Entry:

Sales	900,000	
Cost of Goods Sold		850,000
Inventory	50,000	



20X8 Upstream Sales: 20X8 Equity Accounts

Investment in		Income from Sub	
Sub			
Low	19,800		
	19,800	X7 Reversal	19,800
90% NI	202,500		202,500 90% NI
	45,000	X8 Deferral	45,000
Low	45,000		177,300



20X7 & 20X8 Upstream Sales: 20X8 Partial Worksheet

	Parent	Sub	DR	CR	Consol- idated
Income Statement					
Sales	840,000	900,000	900,000		840,000
COGS	700,000	675,000		850,000	503,000
				22,000	
Income from Sub	177,300		177,300	Basic	
Gross Profit	317,300	225,000	1,077,300	872,000	337,000
NCI in NI			19,700	Basic	(19,700)
CI in NI	317,300	225,000	1,097,000	872,000	317,300
Balance Sheet					
Inventory	200,000			50,000	150,000
Investment in Sub	Low by 45,000		19,800	Basic X	
NCI in NA of Sub			2,200		2,200



Learning Objective 6-5

Understand and explain additional considerations associated with consolidation.



Additional Considerations

◆ Sale from one subsidiary to another

- Transfers of inventory often occur between companies that are under common control or ownership.
- The eliminating entries are identical to those presented earlier for sales from a subsidiary to its parent.
- The full amount of any unrealized intercompany profit is eliminated, with the profit elimination allocated proportionately against the ownership interests of the selling subsidiary.



Additional Considerations

◆ Costs associated with transfers

- When one affiliate transfers inventory to another, some additional cost is often incurred.
- Such costs should be treated in the same way as if the affiliates were operating divisions of a single company.



Additional Considerations

◆ Lower-of-cost-or-market

- A company might write down inventory purchased from an affiliate under this rule if the market value at the end of the period is less than the intercompany transfer price.



Lower-of-Cost-or-Market Example

Assume that a parent company purchases inventory for \$20,000 and sells it to its subsidiary for \$35,000. The subsidiary still holds the inventory at year-end and determines that its market value (replacement cost) is \$25,000 at that time. The subsidiary writes the inventory down from \$35,000 to its lower market value of \$25,000 at the end of the year and records the following entry:

Write-down Inventory to Market Value:

Loss on Decline in Value of Inventory	10,000
Inventory	10,000

Make the following worksheet eliminating entry:

Sales	35,000		
Cost of Goods sold		20,000	
Inventory		5,000	
Loss on Decline in Value of Inventory			10,000



Additional Considerations

◆ Sales and purchases before affiliation

- The consolidation treatment of profits on inventory transfers that occurred before the business combination depends on whether the companies were at that time independent and the sale transaction was the result of arm's-length bargaining.
- As a general rule, the effects of transactions that are not the result of arm's-length bargaining must be eliminated.



Additional Considerations

- ◆ In the absence of evidence to the contrary, companies that have joined together in a business combination are viewed as having been separate and independent prior to the combination.
 - If the prior sales were the result of arm's-length bargaining, they are viewed as transactions between unrelated parties.
 - No elimination or adjustment is needed in preparing consolidated statements subsequent to the combination, even if an affiliate still holds the inventory.



Practice Quiz Question #6

Peanut Co. regularly purchased inventory from Snack Inc. in 20X3 when Peanut did not own any Snack stock. On March 31, 20X4, Peanut purchased 90% of Snack Inc.'s outstanding common stock.

- a. Peanut should eliminate 90% of Snack's first quarter 20X4 gross profit.
- b. Peanut should eliminate 100% of Snack's first quarter 20X4 gross profit.
- c. Peanut should not eliminate any of Snack's first quarter 20X4 gross profit.
- d. Peanut should eliminate 100% of Snack's



Practice Quiz Question #6 *Solution*

Peanut Co. regularly purchased inventory from Snack Inc. in 20X3 when Peanut did not own any Snack stock. On March 31, 20X4, Peanut purchased 90% of Snack Inc.'s outstanding common stock.

- a. Peanut should eliminate 90% of Snack's first quarter 20X4 gross profit.
- b. Peanut should eliminate 100% of Snack's first quarter 20X4 gross profit.**
- c. Peanut should not eliminate any of Snack's first quarter 20X4 gross profit.
- d. Peanut should eliminate 100% of Snack's



quiz

- ◆ Housing Corp., which sells a broad line of home appliances, owns 75 percent of the stock of Wood Company. During 20X8, Wood sold furniture to Housing for \$180,000, which it had produced for \$120,000. Housing sold \$150,000 of its purchase from Wood in 20X8 and the remainder in 20X9. In addition, Housing purchased \$240,000 of inventory from Wood in 20X9 and resold \$90,000 of the items before year-end. Wood's cost to produce the items sold to Housing in 20X9 was \$160,000.



quiz

a. give all elimination entries needed for December 31, 20X9 to remove the effects of the intercompany inventory transfers in 20X8 and 20X9

20X8 Sale:

	<u>Total</u>	=	<u>Re-sold</u>	+	<u>Ending Inventory</u>
Sales	180,000		150,000		30,000
COGS	120,000		100,000		20,000
Gross Profit	60,000		50,000		10,000
Gross Profit %	33.33%				

20X9 Sale:

	<u>Total</u>	=	<u>Re-sold</u>	+	<u>Ending Inventory</u>
Sales	240,000		90,000		150,000
COGS	160,000		60,000		100,000
Gross Profit	80,000		30,000		50,000
Gross Profit %	33.33%				

Investment in Level Brothers	7,500	
NCI in NA of Level Brothers	2,500	
Cost of goods sold		10,000

Reversal of 20X8 gross profit deferral

Sales	240,000	
Cost of Goods Sold		190,000
Inventory		50,000

Eliminate 20X9 intercompany sale of inventory.



quiz

b. compute the amount of income assigned to noncontrolling shareholders in the 20X8 and 20X9 consolidated income statement if Wood reported net income of \$350,000 for 20X8 and \$420,000 for 20X9

	<u>20X8</u>	<u>20X9</u>
Reported net income of Level Brothers	\$350,000	\$420,000
Unrealized profit, December 31, 20X8	(10,000)	10,000
Unrealized profit, December 31, 20X9		<u>(50,000)</u>
Realized net income	\$340,000	\$380,000
Noncontrolling interest's share of ownership	x 0.25	x 0.25
Income assigned to noncontrolling interest	<u>\$ 85,000</u>	<u>\$ 95,000</u>



The End